Nametable Mapping

File: nametable_mapping.kicad_sch

CPU Address Bus

File: cpuAddressBus.kicad_sch

Register Access

File: register_access.kicad_sch

PRG Banking

File: prg_banking.kicad_sch

PPU Address Bus

File: ppu_address_bus.kicad_sch

Reset Detection

File: reset_detection.kicad_sch

CPU Data Bus

File: cpu_data_bus.kicad_sch

PCM DAC

File: pcm_dac.kicad_sch

Scanline Detection

File: scanline_detection.kicad_sch

V-Split

File: v_split.kicad_sch

PPU Fetch Counting

File: ppu_fetch_counting.kicad_sch

CHR A0,A1,A2

File: chr_a0_a1_a2.kicad_sch

ExRAM Mode

File: exram_mode.kicad_sch

Fill Mode

File: fill_mode.kicad_sch

Pulse Length Lookup Table

File: pulse_length_lookup_table.kicad_sch

Pulse Timer

File: pulse_timer.kicad_sch

Pulse Duty Envelope Volume

File: pulse_duty_envelope_volume.kicad_sch

Pulse DAC

File: Pulse DAC.kicad_sch

PPU Data Bus

File: ppu_data_bus.kicad_sch

CHR Bank Registers

File: chr_bank_registers.kicad_sch

CHR Bank Selection Logic

File: chr_bank_selection_logic.kicad_sch

CHR Address Bus 10 to 14

File: chr_address_bus_10_to_14.kicad_sch

CHR Address Bus 15 to 19

File: chr_address_bus_15_to_19.kicad_sch

Multiplier

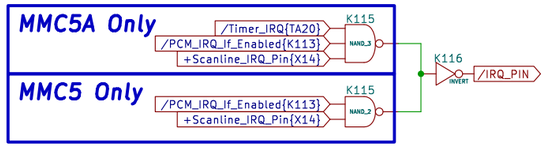
File: multiplier.kicad_sch

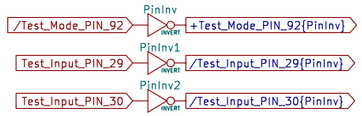
MMC5A Timer

File: mmc5a_timer.kicad_sch

Artifacts

File: artifacts.kicad_sch



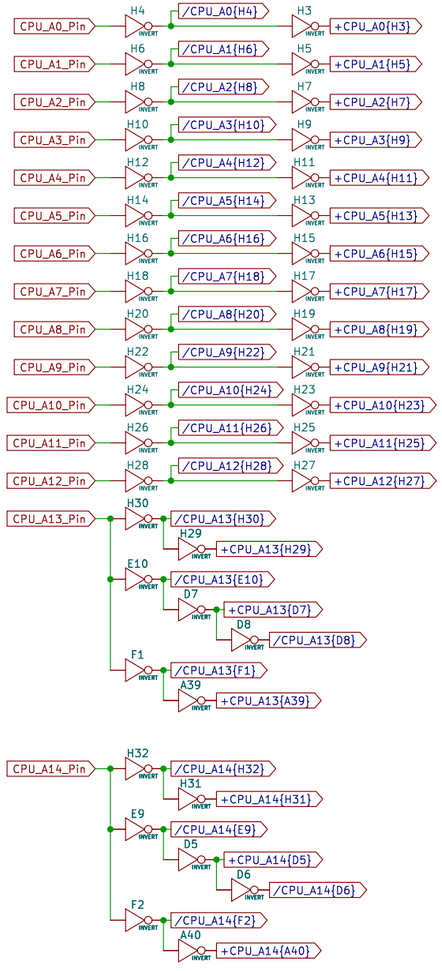


Sheet: /
File: MMC5.kicad_sch

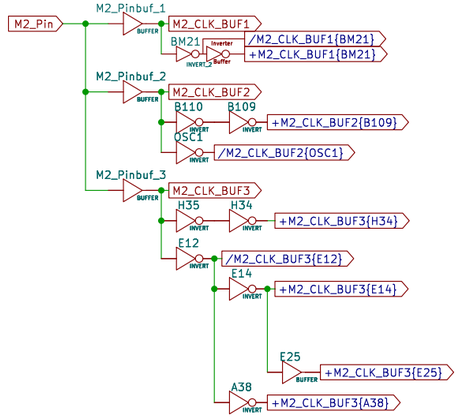
Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 1/27

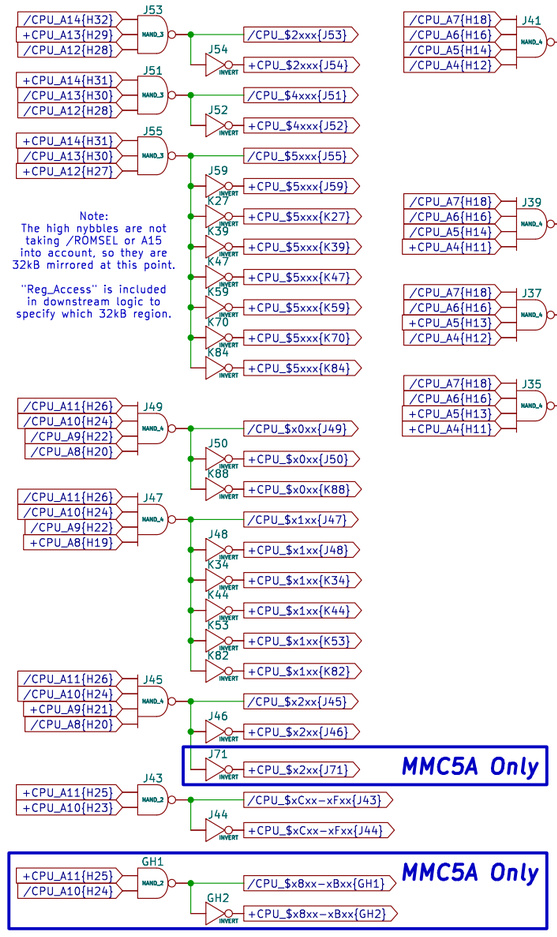
CPU Address Bus Buffering



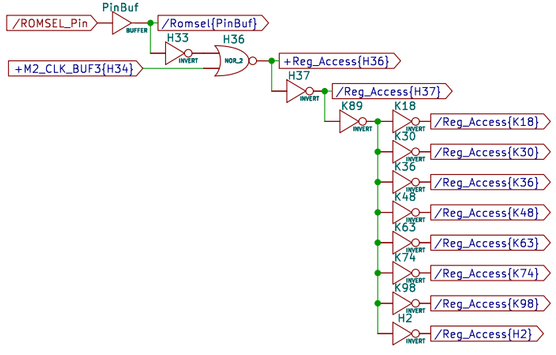
M2 Clock Buffering



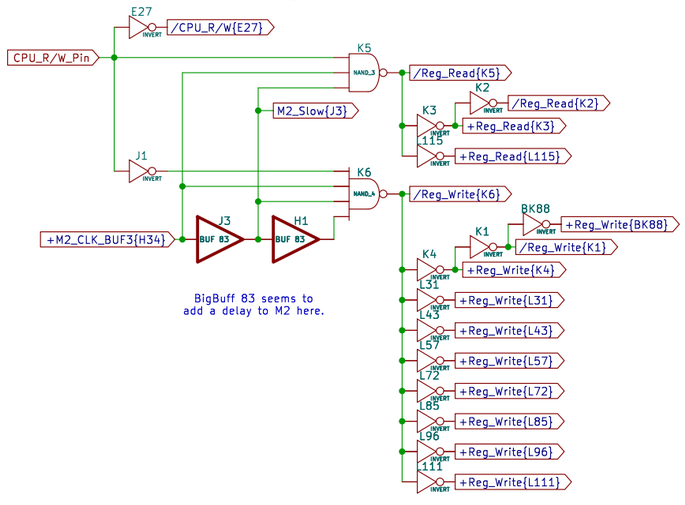
CPU Address Bus Nybble Decoding



/ROMSEL, Register Access



CPU R/W, Register Read/Write

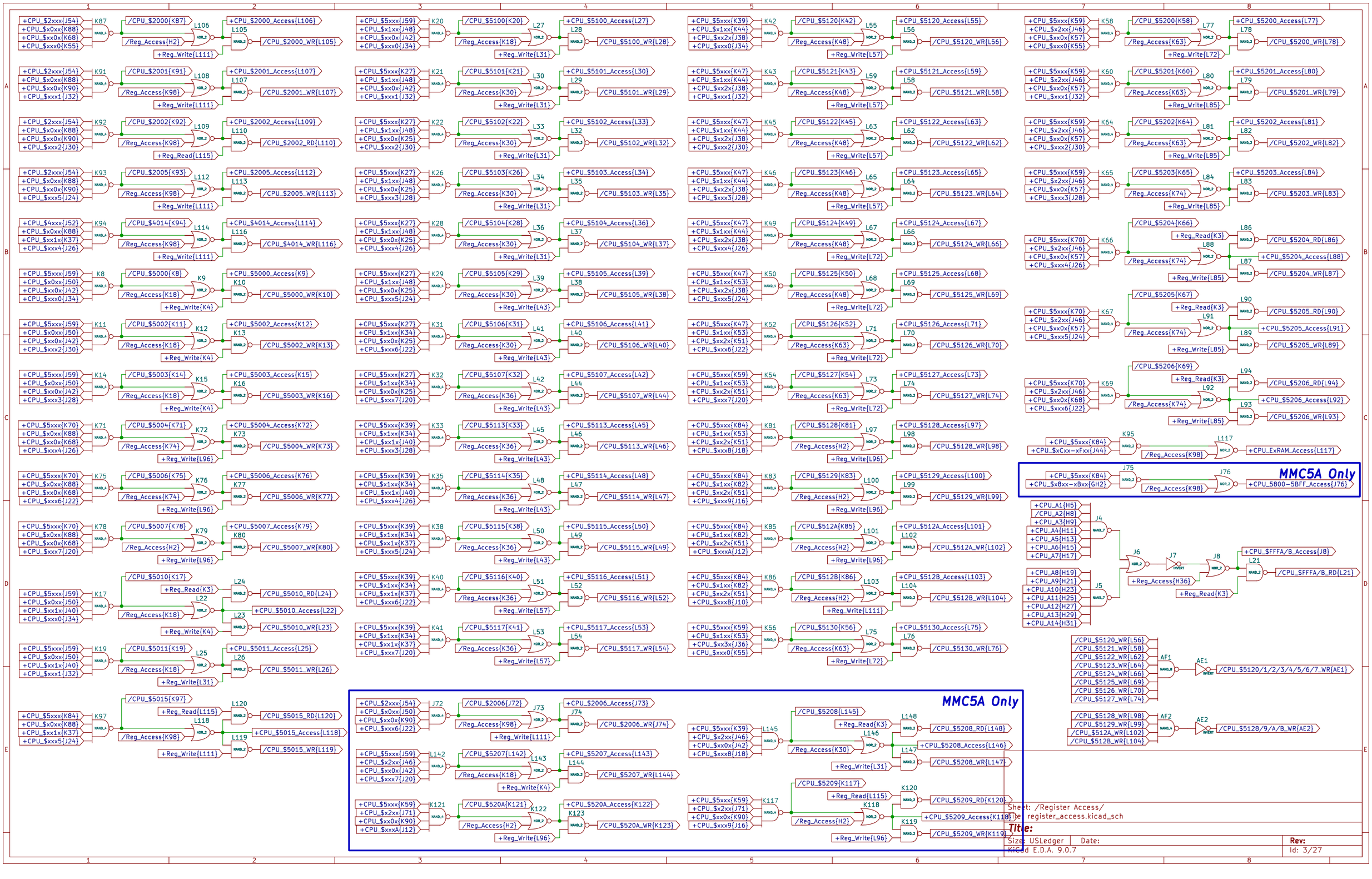


Sheet: /CPU Address Bus/
File: cpuAddressBus.kicad_sch

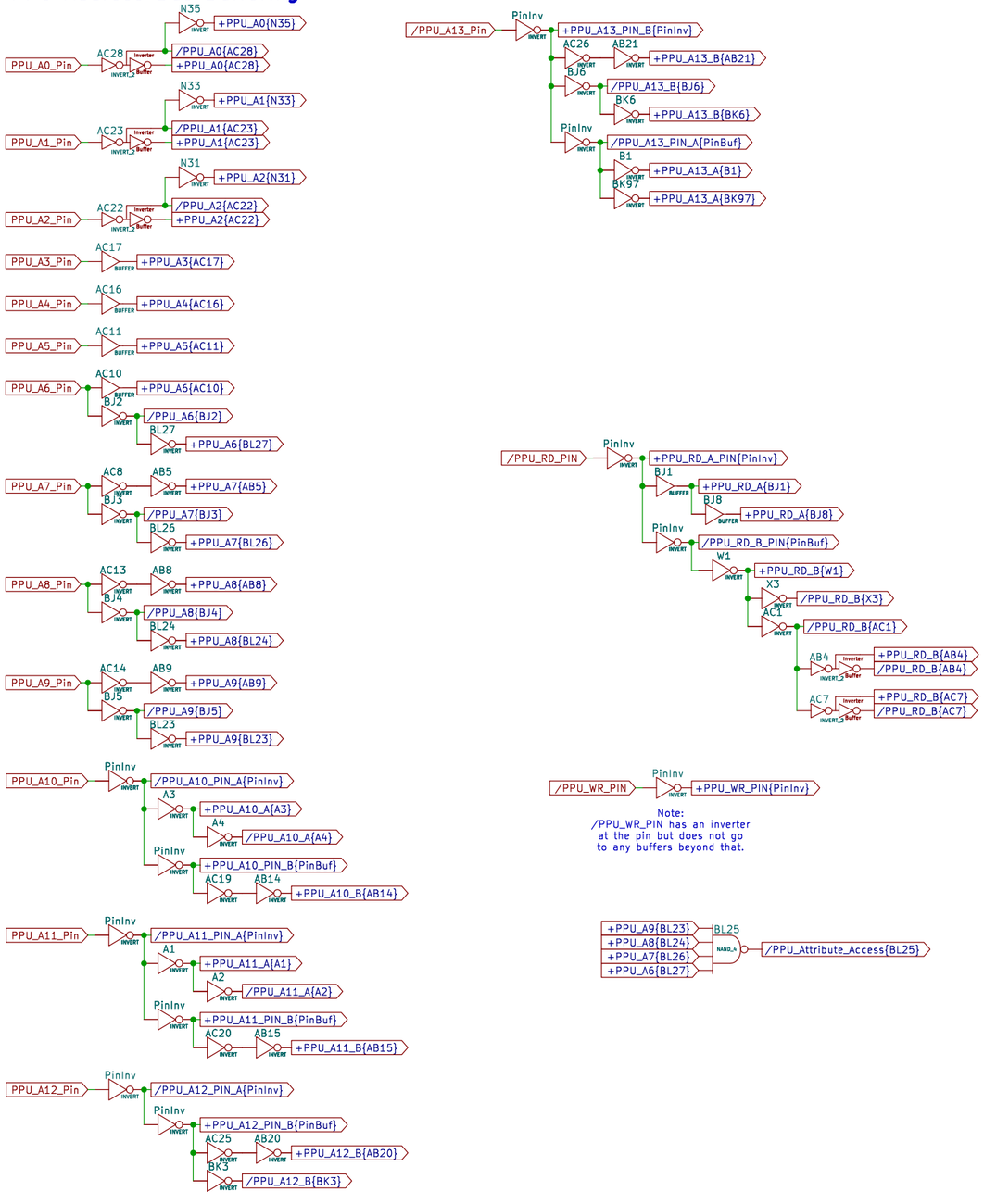
Title:

Size: USLedger | Date: | **Rev:**

KiCad E.D.A. 9.0.7 | Id: 2/27

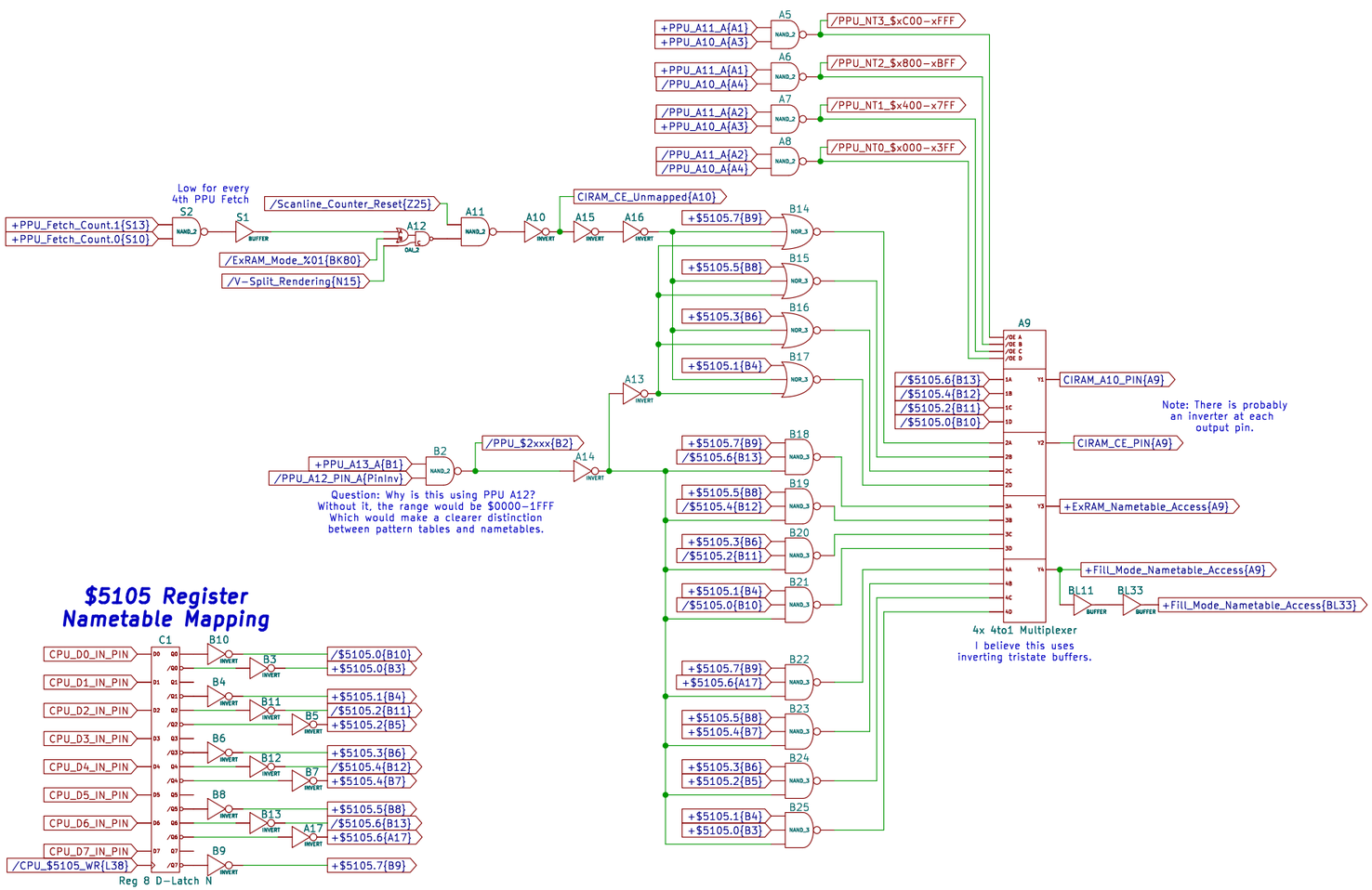

MMC5A Only
MMC5A Only
Sheet: /Register Access/
File: register_access.kicad_sch
Title:
Size: USLedger
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 3/27

PPU Address Bus Buffering



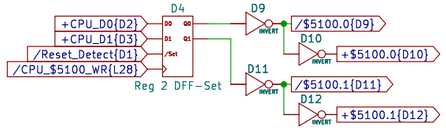
Sheet: /PPU Address Bus/	
File: ppu_address_bus.kicad_sch	
Title:	
Size: USLedger	Date:
KiCad E.D.A. 9.0.7	Id: 4/27

$5105 Register Nametable Mapping

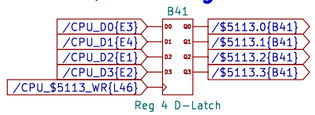


Sheet: /Nametable Mapping/		
File: nametable_mapping.kicad_sch		
Title:		
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 5/27

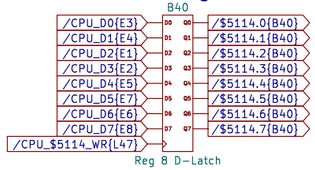
$5100 Register
PRG Mode



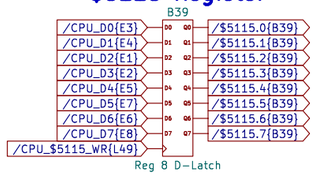
PRG Bank Registers
$5113 Register



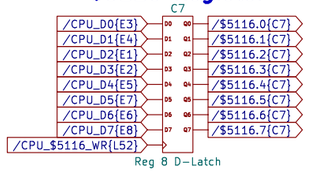
$5114 Register



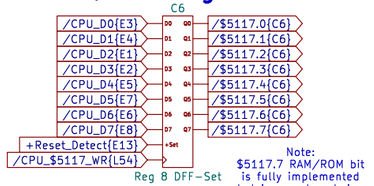
$5115 Register



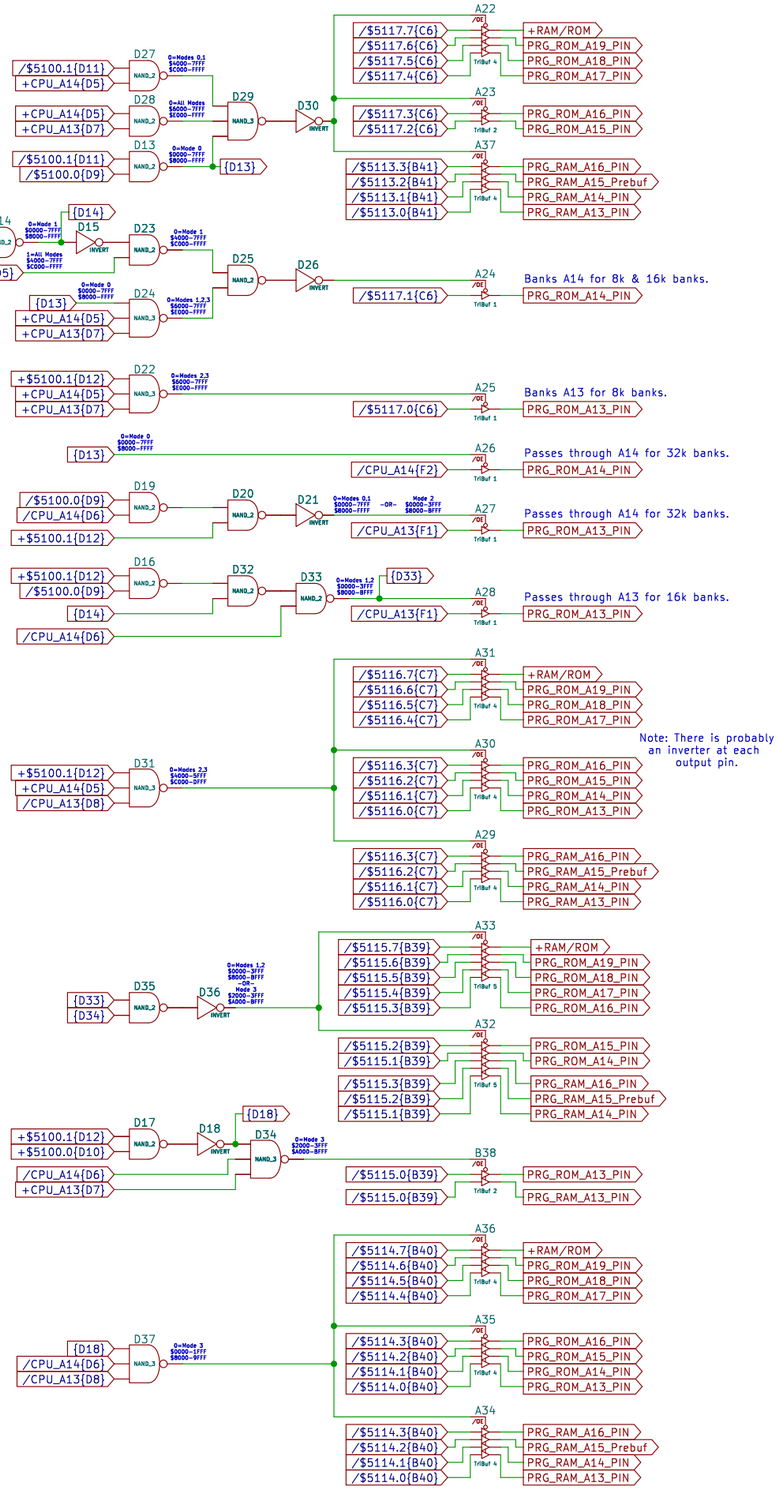
$5116 Register

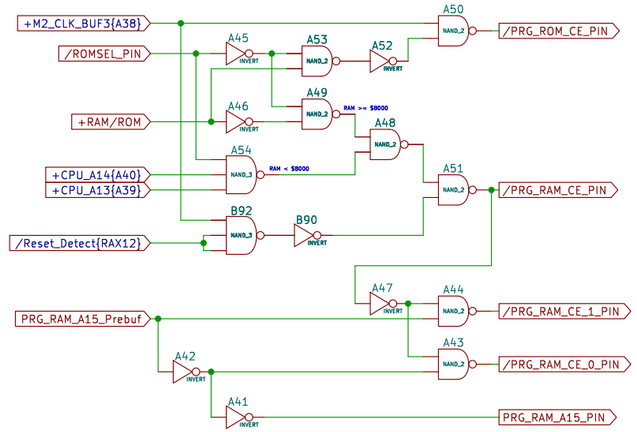


$5117 Register

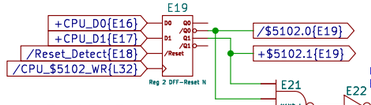


Note: $5117.7 RAM/ROM bit is fully implemented but is somehow being prevented from entering RAM mode when bench testing.

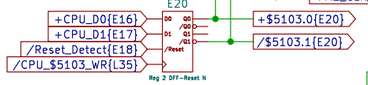




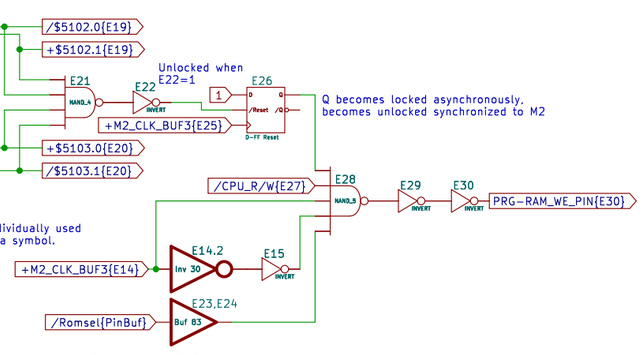
$5102 Register
PRG-RAM Write Protection



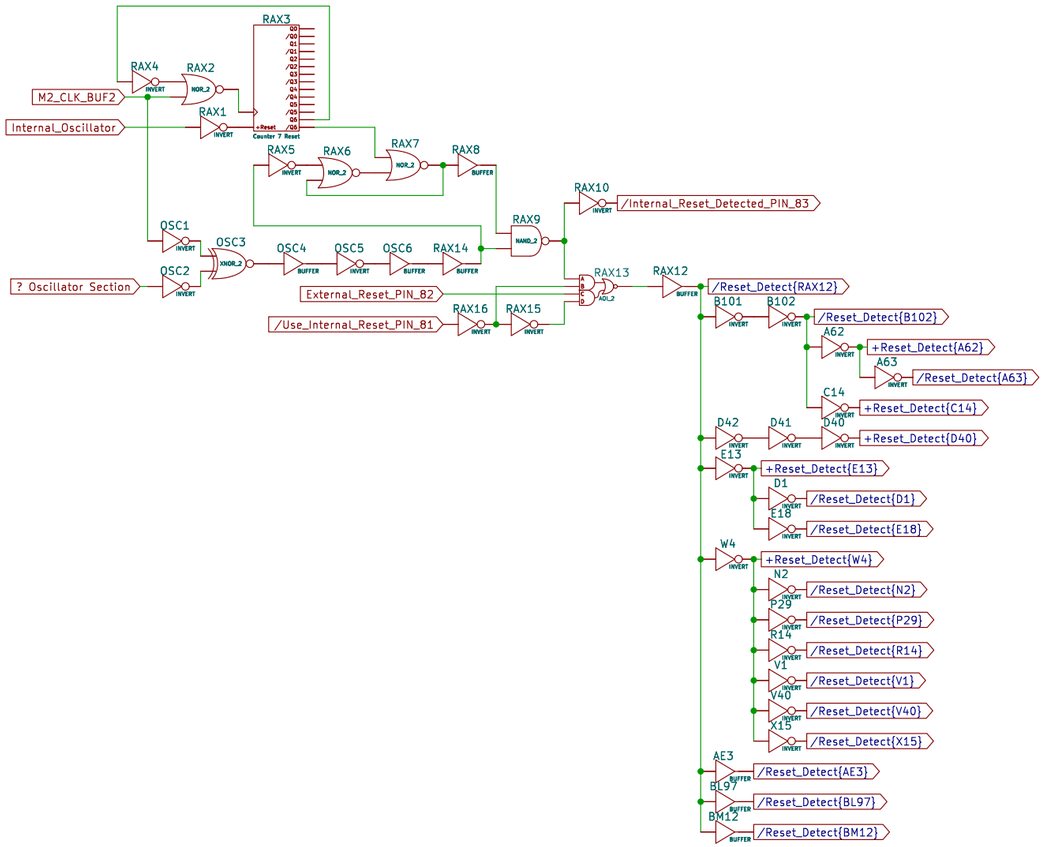
$5103 Register

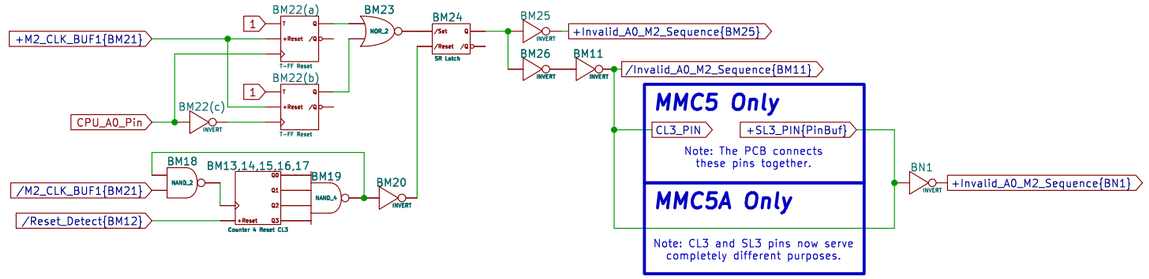


Note: /Set vs. /Reset inputs are individually used but not easily shown here with a symbol.



Sheet: /PRG Banking/	
File: prg_banking.kicad_sch	
Title:	
Size: USLedger Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 6/27





Sheet: /Reset Detection/
File: reset_detection.kicad_sch

Title:

Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 7/27

Incoming CPU Data Bus Buffering Bonanza

CPU_D0_IN_PIN → E3 → /CPU_D0{E3}
- D2 → +CPU_D0{D2}
- E16 → +CPU_D0{E16}
- G15 → G16 → +CPU_D0{G16}
  - L122 → /CPU_D0{L122}
- X18 → /CPU_D0{X18}
  - P26 → +CPU_D0{P26}
  - Y4 → +CPU_D0{Y4}
- AE16 → /CPU_D0{AE16}
- AE18 → /CPU_D0{AE18}
- MUL19 → +CPU_D0{MUL19}
- TF1 → +CPU_D0{TF1}

MMC5A Only

CPU_D1_IN_PIN → E4 → /CPU_D1{E4}
- D3 → +CPU_D1{D3}
- E17 → +CPU_D1{E17}
- G13 → G14 → +CPU_D1{G14}
  - L123 → /CPU_D1{L123}
- X19 → /CPU_D1{X19}
  - P19 → +CPU_D1{P19}
  - Y5 → +CPU_D1{Y5}
- AE15 → /CPU_D1{AE15}
- AE19 → /CPU_D1{AE19}
- MUL20 → +CPU_D1{MUL20}
- TF2 → +CPU_D1{TF2}

MMC5A Only

CPU_D2_IN_PIN → E1 → /CPU_D2{E1}
- G11 → G12 → +CPU_D2{G12}
  - L124 → /CPU_D2{L124}
- X20 → /CPU_D2{X20}
  - P13 → +CPU_D2{P13}
  - Y11 → +CPU_D2{Y11}
- AE20 → /CPU_D2{AE20}
- MUL21 → +CPU_D2{MUL21}
- TF3 → +CPU_D2{TF3}

MMC5A Only

CPU_D3_IN_PIN → E2 → /CPU_D3{E2}
- G9 → G10 → +CPU_D3{G10}
  - E55 → /CPU_D3{E55}
    - E60 → +CPU_D3{E60}
  - L125 → /CPU_D3{L125}
- X21 → /CPU_D3{X21}
  - Y12 → +CPU_D3{Y12}
- AE25 → /CPU_D3{AE25}
- MUL22 → +CPU_D3{MUL22}
- TF4 → +CPU_D3{TF4}

MMC5A Only

CPU_D4_IN_PIN → E5 → /CPU_D4{E5}
- X22 → /CPU_D4{X22}
  - Y18 → +CPU_D4{Y18}
- G7 → G8 → +CPU_D4{G8}
  - E56 → /CPU_D4{E56}
    - E61 → +CPU_D4{E61}
  - L126 → /CPU_D4{L126}
- AE24 → /CPU_D4{AE24}
- MUL23 → +CPU_D4{MUL23}
- TF5 → +CPU_D4{TF5}

MMC5A Only

CPU_D5_IN_PIN → E7 → /CPU_D5{E7}
- G5 → G6 → +CPU_D5{G6}
  - E57 → /CPU_D5{E57}
    - E62 → +CPU_D5{E62}
  - L127 → /CPU_D5{L127}
- X23 → /CPU_D5{X23}
  - Y19 → +CPU_D5{Y19}
- AE23 → /CPU_D5{AE23}
- MUL24 → +CPU_D5{MUL24}
- TF6 → +CPU_D5{TF6}

MMC5A Only

CPU_D6_IN_PIN → E6 → /CPU_D6{E6}
- G3 → G4 → +CPU_D6{G4}
  - E58 → /CPU_D6{E58}
    - E63 → +CPU_D6{E63}
  - L128 → /CPU_D6{L128}
- X24 → /CPU_D6{X24}
  - P7 → +CPU_D6{P7}
  - Y25 → +CPU_D6{Y25}
- AE22 → /CPU_D6{AE22}
- MUL25 → +CPU_D6{MUL25}
- TF7 → +CPU_D6{TF7}

MMC5A Only

CPU_D7_IN_PIN → E8 → /CPU_D7{E8}
- G1 → G2 → +CPU_D7{G2}
  - E59 → /CPU_D7{E59}
    - E64 → +CPU_D7{E64}
  - L129 → /CPU_D7{L129}
- X25 → /CPU_D7{X25}
  - P5 → +CPU_D7{P5}
  - Y26 → +CPU_D7{Y26}
- AE21 → /CPU_D7{AE21}
- MUL26 → +CPU_D7{MUL26}
- TF8 → +CPU_D7{TF8}

MMC5A Only

Status Bit Buffers

$5015.0_Read_Pulse_1_Status_Bit{B106}	BL21	$5015.0_Read_Pulse_1_Status_Bit{BL21}
$5015.1_Read_Pulse_2_Status_Bit{B104}	BL20	$5015.1_Read_Pulse_2_Status_Bit{BL20}
+Invalid_A0_M2_Sequence{BM25}	BL52	$5204.0_Read_Invalid_A0_M2_Sequence{BL52}
+PCM_IRQ_Status_Bit{K114}	BL62	$5010.7_Read_PCM_IRQ_Pending{BL62}
/Scanline_IRQ_Status{X28}	BL79	$5204.7_Read_Scanline_IRQ_Pending{BL79}
+In_Frame{Z21}	BL80	$5010.6_Read_In_Frame{BL80}

MMC5A Only

Pin_97_CL3_GPIO_In	BM0.7	$5208.6_CL3_GPIO_In{BM0.7}
Pin_98_SL3_GPIO_In	BM0.6	$5208.7_SL3_GPIO_In{BM0.6}
+Timer_IRQ_Status_Bit{TA22}	BM0.4	$5209.7_Timer_IRQ_Status_Bit{BM0.4}

Decides When to Drive the CPU Data Bus

MMC5A Only

+CPU_$5208_Access{L146}	BM0.2	+CPU_$5208_Access{BM0.2}	BP0
+CPU_$5209_Access{K118}	BM0.1	+CPU_$5209_Access{BM0.1}	BK54.5A

→ BP2 (NAND_6)

MMC5 Only

+CPU_$5015_Access{L118}	BL22	+CPU_$5015_Access{BL22}	BP1
+CPU_$5204_Access{L88}	BL56	+CPU_$5204_Access{BL56}	BK48
+CPU_$5205_Access{L91}	BL55	+CPU_$5205_Access{BL55}	BP3
+CPU_$5206_Access{L92}	BL54	+CPU_$5206_Access{BL54}	BP4
+CPU_ExRAM_Access{L117}	BL51	+CPU_ExRAM_Access{BL51}	BK52
+ExRAM_Mode_$0x{BL96}	BL48	+ExRAM_Mode_$0x{BL48}	BK50
+CPU_$5010_Access{L22}	BL53	+CPU_$5010_Access{BL53}	BK49
/Reg_Read{K2}	BL50		BK47

BP2 (NAND_4) → BK51 (OAI_2) → BK46 (OAI_2) → BK45 (INVERT) → CPU_Dx_Direction{BK45}

Multiplexing Data to be Driven onto the CPU Data Bus

I wonder why they used _RD instead of _Access for the new status bits? They even went so far as to run $5010_RD way across the die when $5010_Access was already here.

CPU_Dx_Direction{BK45} already considers Reg_Read and its applicable /ROMSEL delay.

+CPU_$5206_Access{BL54} → BL58
+CPU_$5205_Access{BL55} → BL59
+CPU_$5204_Access{BL56} → BL60
+CPU_$5010_Access{BL53} → BL57

MMC5A Only

/CPU_$5010_RD{L24}	BM0.8	BN0.6
/CPU_$5208_RD{L148}	BM0.5	BN0.3
/CPU_$5209_RD{K120}	BM0.3	BN0.1

+CPU_$5015_Access{BL22}

+CPU_ExRAM_Access{BL51} → BL61.1 (INVERT)
+ExRAM_Mode_$0x{BL48} → BL61.2 (NOR_2)

ExRAM_D0_Read{BJ12(a)} → BK53 (NAND_3)
$5206.0_Read_Multiply_Result_D8{BL78}
$5204.0_Read_Invalid_A0_M2_Sequence{BL52}
$5205.0_Read_Multiply_Result_D0{BL70}
$5015.0_Read_Pulse_1_Status_Bit{BL21}
→ BK55 (ADL_24) → BK54 (NAND_2) → CPU_D0_OUT_PIN

MMC5 Only

$5010.0 MMC5A Revision Bit: 1 → BN0.5 (NAND_2) → BR3 (NAND_3)

MMC5A Only

ExRAM_D1_Read{BJ12(b)} → BK58 (NAND_2) → BK57 → CPU_D1_OUT_PIN
$5206.1_Read_Multiply_Result_D9{BL77}
$5205.1_Read_Multiply_Result_D1{BL69}
$5015.1_Read_Pulse_2_Status_Bit{BL20}
→ BK56 (ADL_23)

ExRAM_D2_Read{BJ12(c)} → BK59 (NAND_2) → BK60 → CPU_D2_OUT_PIN
$5206.2_Read_Multiply_Result_D10{BL76}
$5205.2_Read_Multiply_Result_D2{BL68}
→ BK61 (ADL_2)

ExRAM_D3_Read{BJ12(d)} → BK64 (NAND_2) → BK63 → CPU_D3_OUT_PIN
$5206.3_Read_Multiply_Result_D11{BL75}
$5205.3_Read_Multiply_Result_D3{BL67}
→ BK62 (ADL_2)

ExRAM_D4_Read{BJ12(e)} → BK65 (NAND_2) → BK66 → CPU_D4_OUT_PIN
$5206.4_Read_Multiply_Result_D12{BL74}
$5205.4_Read_Multiply_Result_D4{BL66}
→ BK67 (ADL_2)

ExRAM_D5_Read{BJ12(f)} → BK70 (NAND_2) → BK69 → CPU_D5_OUT_PIN
$5206.5_Read_Multiply_Result_D13{BL73}
$5205.5_Read_Multiply_Result_D5{BL65}
→ BK68 (ADL_2)

ExRAM_D6_Read{BJ12(g)} → BK71 (NAND_2) → BK72 → CPU_D6_OUT_PIN
$5206.6_Read_Multiply_Result_D14{BL64}
$5205.6_Read_Multiply_Result_D6{BL72}
$5010.6_Read_In_Frame{BL80}
→ BK73 (ADL_23)

MMC5 Only

$5208.6_CL3_GPIO_In{BM0.7} → BN0.4 (NAND_2) → BR2 (NAND_3)

MMC5A Only

ExRAM_D7_Read{BJ12(h)} → BK76 (NAND_2) → BK75 → CPU_D7_OUT_PIN
$5206.7_Read_Multiply_Result_D15{BL63}
$5205.7_Read_Multiply_Result_D7{BL71}
$5204.7_Read_Scanline_IRQ_Pending{BL79}
$5010.7_Read_PCM_IRQ_Pending{BL62}
→ BK74 (ADL_24)

MMC5 Only

$5209.7_Timer_IRQ_Status_Bit{BM0.4} → BN0.2 (NAND_2) → BR1 (NAND_3)
$5208.7_SL3_GPIO_In{BM0.6}

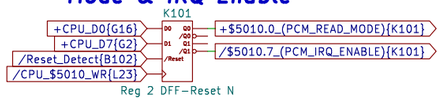
MMC5A Only

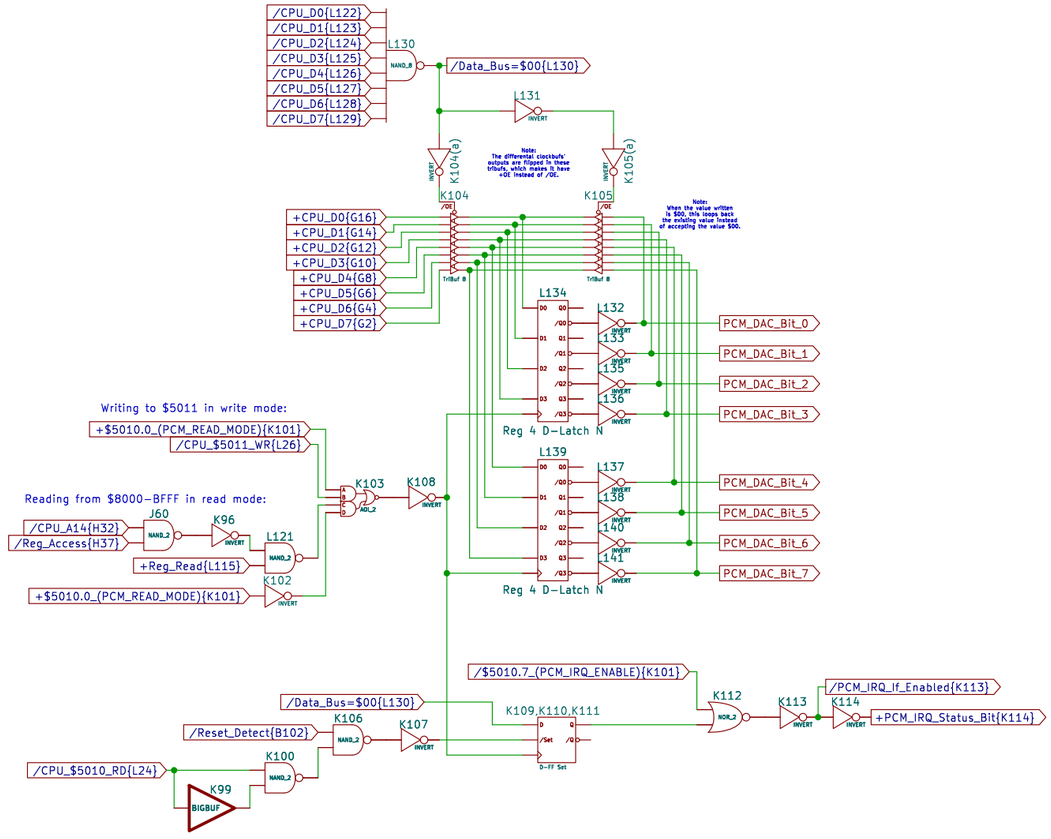
Sheet: /CPU Data Bus/
File: cpu_data_bus.kicad_sch

Title:

Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 8/27

$5010 Register
Mode & IRQ Enable





Sheet: /PCM DAC/	
File: pcm_dac.kicad_sch	
Title:	
Size: USLedger | Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 9/27

Scanline Start Detection

+Use Internal Scanline Detection

+Use External Scanline Detection

+Test_Mode_PIN_92{Pininv} → AA1 (INVERT,3 Inverter Buffer)

/Test_Input_PIN_29{Pininv} → AA4 (INVERT)

External Scanline Start

High when the present PPU address is from $2xxx and matches the previous read.

AA5 (NOR_4) ← +PPU_A13_B{AB21}, +PPU_A12_B{AB20}, +PPU_A11_B{AB15}, +PPU_A10_B{AB14}, +PPU_A9{AB9}, +PPU_A8{AB8}, +PPU_A7{AB5}, +PPU_A6{AC10}, +PPU_A5{AC11}, +PPU_A4{AC16}, +PPU_A3{AC17}, +PPU_A2{AC22}, +PPU_A1{AC23}, +PPU_A0{AC28}

+PPU_A13_B{AB21} → AB22 → AA21 (NAND_2) → AA20 (INVERT)

+PPU_A12_B{AB20} → AB19 → AA18 (NOR_2)

+PPU_A11_B{AB15} → AB16 → AA17 (XNOR_2)

+PPU_A10_B{AB14} → AB13 → AA14 (XNOR_2)

+PPU_A9{AB9} → AB10 → AA13 (XNOR_2)

+PPU_A8{AB8} → AB7 → AA12 (XNOR_2)

+PPU_A7{AB5} → AB6 → AA9 (XNOR_2)

+PPU_A6{AC10} → AC9 → AA8 (XNOR_2)

+PPU_A5{AC11} → AC12 → AA11 (XNOR_2)

+PPU_A4{AC16} → AC15 → AB11 (XNOR_2)

+PPU_A3{AC17} → AC18 → AB12 (XNOR_2)

+PPU_A2{AC22} → AC21 → AB17 (XNOR_2)

+PPU_A1{AC23} → AC24 → AB18 (XNOR_2)

+PPU_A0{AC28} → AC27 → AA22 (XNOR_2)

AA16 (NAND_4) ← AA14, AA13, AA12, AA9

AA10 (NAND_4) ← AA18, AA17, AA8, AA11

AA15 (NAND_4) ← AB11, AB12, AB17, AB18

AA19 (NAND_2) ← AA16, AA10, AA15

/PPU_RD_B{AB4} → AA7 (INVERT)

/Exit_In_Frame{AC6} → AA6 (INVERT)

/PPU_RD_B{AC7} → AB3 (INVERT)

When not in frame, the previous PPU read address always looks like $0000. Since A13, A12 are looking at range $2xxx instead of matches, $0xxx won't match $2xxx. Therefore, scanlines don't get detected when using PPUDATA to read from $0000 during v-blank.

AB2 (SR Reg 3) ← +PPU_RD_B{W1}, AA5 (NOR_4)

AB2(a) (INVERT)

AB1 (NAND_4) ← AB2, AB2(a)

Low when the 3 most recent PPU reads were from $2xxx and all matched.

AA2 (AOI_2) ← AB1, +Use Internal Scanline Detection, +Use External Scanline Detection

AA3 (INVERT) → /Scanline_Start_Detected{AA3}

R2 (BUFFER) → /Scanline_Start_Detected{R2}

AC2 (Counter 2 Reset) ← M2_CLK_BUF1, /PPU_RD_B{AC1}

AC6 (NAND_2) ← AC2 → /Exit_In_Frame{AC6}

Scanline End Detection

Increment_Scanline_Fetch_Count{S6}, +PPU_Cycle_336{R4} → R6 (NAND_2) → R7 (INVERT) → R8 (INVERT) → Scanline_End_Detected_Internal{R8}

Scanline_End_Detected_Internal{R8}, /Test_Input_PIN_30{Pininv} (P4 INVERT) External Scanline End → R1 (AOI_3) → P3 (INVERT) → Scanline_End_Detected{P3}

P3 → T5 (BUFFER) → Scanline_End_Detected{T5}

$5204 Register Scanline IRQ Enable

+CPU_D7{Y26}, /Reset_Detect{X15}, /CPU_$5204_WR{L87} → X12 (D-FF Reset)

X12 → X13 (NAND_2) → X14 (BUFFER) → +Scanline_IRQ_Pin{X14}

This signal likely needs to be active low.

$5203 Register IRQ Scanline Compare Value

Y2,7,9,14,16,21,23,28

+CPU_D0{Y4}, +CPU_D1{Y5}, +CPU_D2{Y11}, +CPU_D3{Y12}, +CPU_D4{Y18}, +CPU_D5{Y19}, +CPU_D6{Y25}, +CPU_D7{Y26} → Y1 (BUFFER) Reg 8 D-Latch

/CPU_$5203_WR{L83} → Y1

Y3, Y6, Y10, Y13, Y17, Y20, Y24, Y27 (XNOR_2) → Y8 (NAND_4), Y22 (NAND_4) → Y15 (NOR_2)

Hi when $5203 matches the present V-Split scanline.

D'Morgan tells us that we can pretend Y8,Y15,Y22 are all AND gates.

IRQ Scanline Counter

Z2-Z18 (Counter 8 Reset)

/Scanline_Start_Detected{AA3}, +In_Frame{Z21} → Z1 (NAND_2)

/In_Frame{Z23} → Z19 (BUFFER) → Counter 8 Reset

Z22 (NAND_4) ← Z2-Z18 → /Scanline_240_and_up{Z22}

/CPU_$5204_RD{L86} → M3 (BIGBUF) → N38 (NAND_2) → N39 (INVERT) → M2 (NAND_2)

+New_Scanline_Detected{AA3 via Z23}, /In_Frame{Z23} → Z20 (NAND_2)

M2, Z20 → X17 (NAND_2) → X26 (INVERT) → Y31 (D-FF Reset)

/Scanline_Start_Detected{AA3} → Y31

Y31 → +IRQ → Y29 (NOR_2) → Y30 (INVERT) → /CPU_$5204_RD{L86}

Y31 → +IRQ → Y32 (INVERT) → /IRQ → X27 (D-FF No Reset) → +IRQ → X28 (INVERT) → /Scanline_IRQ_Status{X28}

This signal likely needs to be active hi.

$2001 Register Write Snooping

+CPU_D3{Y12}, +CPU_D4{Y18}, /CPU_$2001_WR{L107} → X8 (Reg 2 DFF-No-Reset N) → X9 (NAND_2)

Hi = PPU Rendering Enabled

X9 → Z24 (NAND_2) → Z25 (INVERT) → +Rendering_In_Frame{Z25}

V39 (INVERT) → /Rendering_In_Frame{V39}

R24 (INVERT) → /Rendering_In_Frame{R24}

N16 (INVERT) → N9 (INVERT) → +Rendering_In_Frame{N9}

$2002 Register Read Snooping

Latches $2002.7 value read by CPU for 1 CPU cycle.

/CPU_$2002_RD{L110} → X16 (INVERT) → X4 (NAND_2)

+CPU_D7{Y26} → X5 (D-FF No Reset)

M2_CLK_BUF1 → X5

/CPU_$FFFA/B_RD{L21}, /CPU_$4014_WR{L116}, /Exit_In_Frame{AC6} → X6 (NAND_4) → X7 (INVERT) → Z23 (D-FF Reset)

/Scanline_Start_Detected{AA3} → Z23

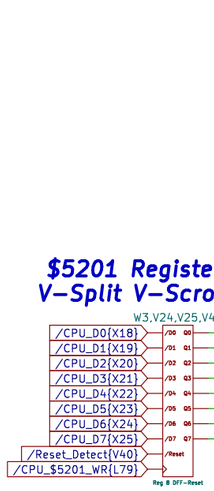
Z23 → /In_Frame{Z23}

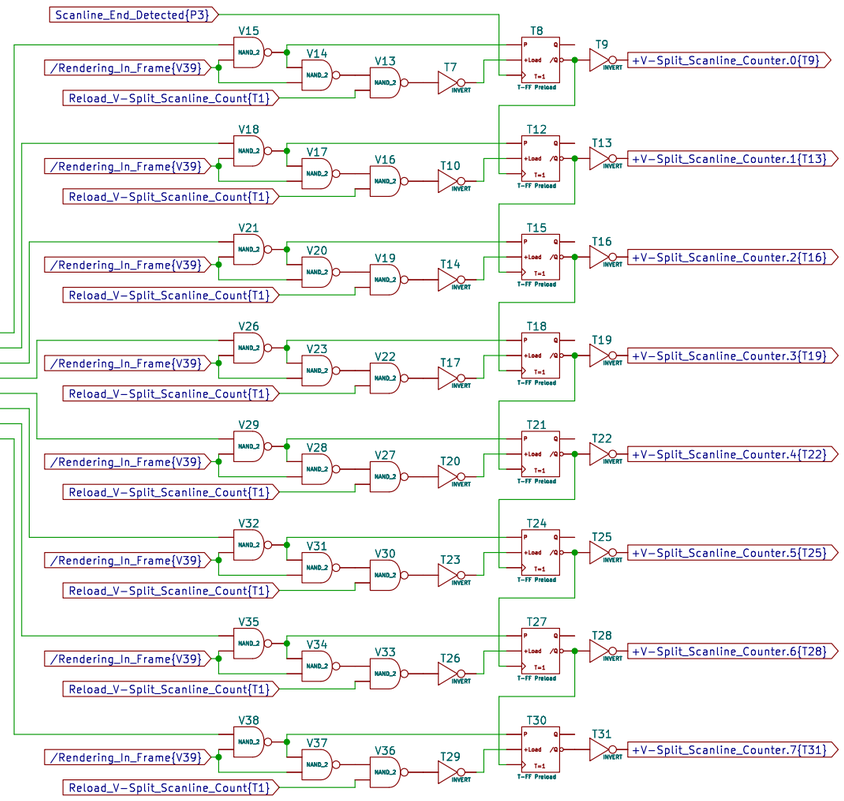
Z23 → Z21 (INVERT) → +In_Frame{Z21}

Sheet: /Scanline Detection/	
File: scanline_detection.kicad_sch	
Title:	
Size: B	Date:
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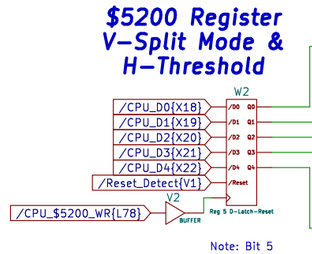
V−Split Scanline Counter

$5201 Register V−Split V−Scroll

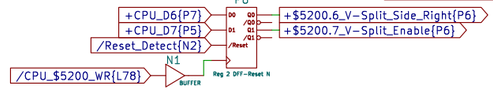


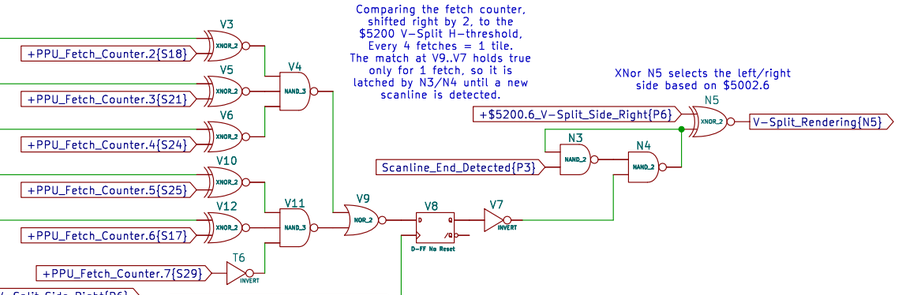


$5200 Register V−Split Mode & H−Threshold



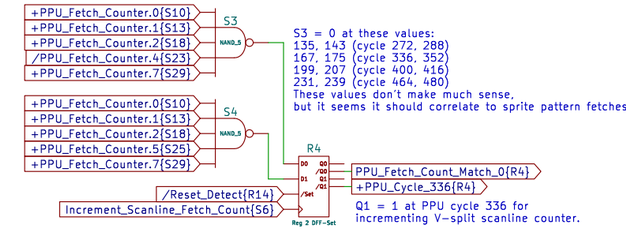
Note: Bit 5 does not exist.





Comparing the fetch counter, shifted right by 2, to the $5200 V−Split H−threshold, Every 4 fetches = 1 tile. The match at V9..V7 holds true only for 1 fetch, so it is latched by N3/N4 until a new scanline is detected.

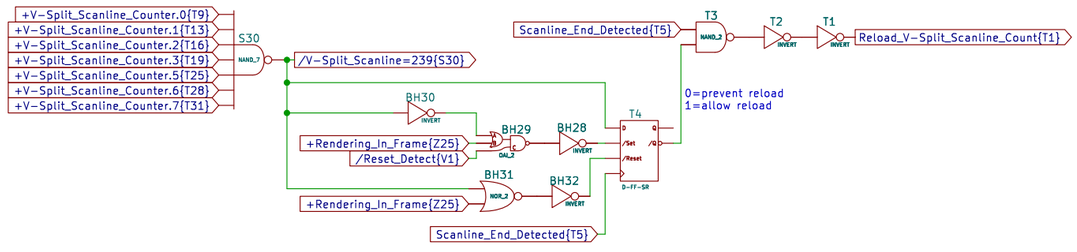
XNor N5 selects the left/right side based on $5002.6



S3 = 0 at these values:
135, 143 (cycle 272, 288)
167, 175 (cycle 336, 352)
199, 207 (cycle 400, 416)
231, 239 (cycle 464, 480)
These values don't make much sense, but it seems it should correlate to sprite pattern fetches

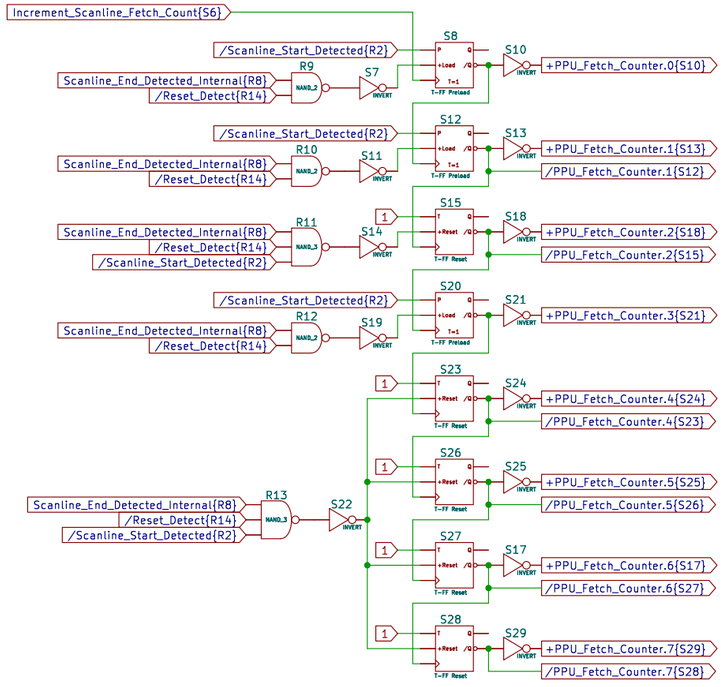
Q1 = 1 at PPU cycle 336 for incrementing V−split scanline counter.

Logic from S4:
Value 1x1xx111, when incrementing,
First triggered at 10100111=0xA7=167.
Because fetch count clocks the latch,
167+1 = 168. Since PPU has 2 cycles per fetch, 168*2=336 cycles.
This aligns with "unused NT fetches" at the end of the scanline.

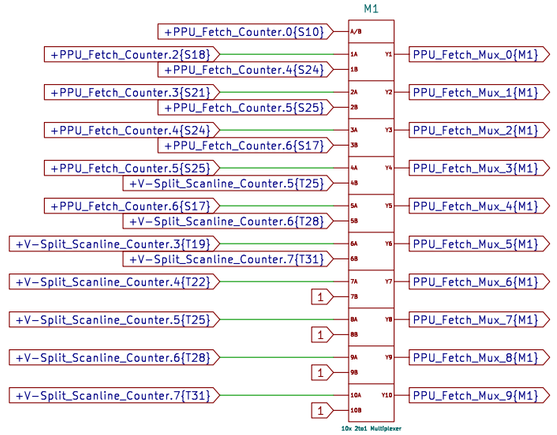


Sheet: /V−Split/		
File: v_split.kicad_sch		
Title:		
Size: B	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 11/27

PPU Fetch Counter

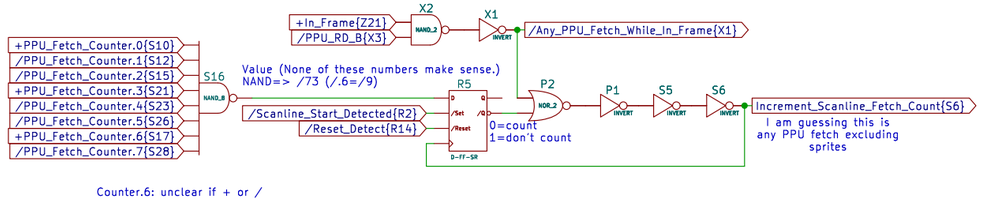


ExRAM Attribute Fetch Mux



The output of this mux is the ExRAM address for the data to be substituted for the PPU.

See sheet 14.



Counter.6: unclear if + or /

Sheet: /PPU Fetch Counting/	
File: ppu_fetch_counting.kicad_sch	
Title:	
Size: USLedger Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 12/27

This area should be verified because
it seems N15 should also go low for
extended attribute fetches.

V-Split_Rendering{N5}
Shift_Q0{R20}
N6
NAND_2
N7
INVERT
+$5200.7_V-Split_Enable{P6}
+ExRAM_Mode_$0x{BL96}
+Rendering_In_Frame{N9}
N8
NAND_A
N10
INVERT
N15
INVERT
BL47
BUFFER
BL44
INVERT
BL43
INVERT
/V-Split_Rendering{BL43}
/V-Split_Rendering{N15}
/VSplit_Pattern_Table_Fetch{N14}
N11,12,13
D
/Set
Q0
Q1
Q2
/Q2
Shift Reg 2 DFF-Set
+Rendering_In_Frame{N9}
/Any_PPU_Fetch_While_In_Frame{X1}
N14
INVERT
N26
INVERT
N22
NAND_2
/OE_A_3-Bit_Counter{N22}
N28
NAND_2
/OE_B_V-Split_Counter{N28}
+Rendering_In_Frame{Z25}
Scanline_End_Detected_Internal{R8}
Increment_Scanline_Fetch_Count{S6}
PPU_Fetch_Count_Match_0{R4}
R3
NAND_2
R15
NAND_2
R16
NAND_2
/Any_PPU_Fetch_While_In_Frame{X1}
R17,18,19,20,21,22,23
D
/Reset
Q0
Q1
Q2
Q3
Shift Reg 3 DFF-Reset
Shift_Q0{R20}
Shift_Q2{R23}
R25
NAND_2
P27
INVERT
/OE_C_PPU_Passthrough{P27}
Theory:
/OE C Active when:
– PPU is fetching sprite pattern data
or not rendering (ex. via PPUDATA).
/OE B Active when:
– PPU is fetching background pattern data
in V-Split region.
/OE A Active when:
– PPU is fetching background pattern data
everywhere else.

/OE_B_V-Split_Counter{N28}

MMC5A Only

N29 is an internal buffer
of the multiplexer.

Never Enables
3-bit Counter
1
+OE_B_V-Split_Counter{N29}

MMC5 Only

/OE_A_3-Bit_Counter{N22}
/OE_C_PPU_Passthrough{P27}

N18,19,20,21,24,25,27,29,32,34,36,37
/OE A
/OE B
/OE C
+3-Bit_Counter.0{P20}
+V-Split_Scanline_Counter.0{T9}
+PPU_A0{N35}
1A
1B
1C
Y1
N30
INVERT
CHR_A0_PIN
+3-Bit_Counter.1{P14}
+V-Split_Scanline_Counter.1{T13}
+PPU_A1{N33}
2A
2B
2C
Y2
N23
INVERT
CHR_A1_PIN
+3-Bit_Counter.2{P8}
+V-Split_Scanline_Counter.2{T16}
+PPU_A2{N31}
3A
3B
3C
Y3
N17
INVERT
CHR_A2_PIN
3x 3to1 Multiplexer

$2006 Register
2nd Write Snooping

+CPU_$2xxx{J54}
+CPU_$x0xx{J50}
+CPU_$xx0x{K90}
+CPU_$xxx6{J22}
J72
NAND_A
/CPU_$2006{J72}
/Reg_Access{K98}
+Reg_Write{L111}
J73
NOR_2
+CPU_$2006_Access{J73}
J74
NAND_2
/CPU_$2006_WR{J74}
DISCH1,2,3
D
/Reset
Q0
Q1
/Q1
Counter 2 Reset
+$2006_2nd_Write{DISCH3}
The counter counts:
0,1,2,3,0,1,2,3...
So Q1 is high for:
2nd, 3rd,
6th, 7th,
10th, 11th...
writes.
DISCH17
?
Oscillator
DISCH16
INVERT
DISCH15
AOI_2
/Scanline_Start_Detected{AA3}
Scanline_End_Detected{P3}
>>>
Theory: this signal oscillates during H-blank,
continuously triggering the 3-bit counter to reload.
/CPU_$2002_RD{L110}
DISCH13
NAND_2
DISCH12
INVERT
DISCH14
NAND_2
DISCH11
BIGBUF
DISCH4
NAND_2
+$2006_2nd_Write{DISCH3}
DISCH9
NAND_2
DISCH10
INVERT
MMC5_3-bit_Reload_0{P23}
MMC5A_3-bit_Reload_0{DISCH10}
3-bit_Reload_0
DISCH7
NAND_2
DISCH8
INVERT
MMC5_3-bit_Reload_1{P16}
MMC5A_3-bit_Reload_1{DISCH8}
3-bit_Reload_1
DISCH5
NAND_2
DISCH6
INVERT
MMC5_3-bit_Reload_2{P10}
MMC5A_3-bit_Reload_2{DISCH6}
3-bit_Reload_2
Seemingly, none of this new stuff
has any effect because the 3-bit
scanline counter can't be enabled
onto CHR A0/A1/A2.

MMC5A Only

MMC5 Only

MMC5_3-bit_Reload_0{P23}
MMC5_3-bit_Reload_1{P16}
MMC5_3-bit_Reload_2{P10}
3-bit_Reload_0
3-bit_Reload_1
3-bit_Reload_2

Fine Vertical Scroll
3-Bit Scanline Counter

$2005 Register
2nd Write Snooping

Scanline_End_Detected{P3}
P22
BUFFER
P24
NAND_2
P23
NAND_2
3-bit_Reload_0
MMC5_3-bit_Reload_0{P23}
P21
P
Q
+Load
/Q
T=1
T-FF Preload
P20
INVERT
+3-Bit_Counter.0{P20}
P12,13,18,19,25,26
/CPU_D0{X18}
/CPU_D1{X19}
/CPU_D2{X20}
/Reset_Detect{P29}
/D0
/D1
/D2
/Reset
Q0
Q1
Q2
Reg 3 DFF-Reset
P17
NAND_2
P16
NAND_2
3-bit_Reload_1
MMC5_3-bit_Reload_1{P16}
P15
P14
+3-Bit_Counter.1{P14}
R26
NAND_2
P28
BUFFER
P11
NAND_2
P10
NAND_2
3-bit_Reload_2
MMC5_3-bit_Reload_2{P10}
P9
P8
+3-Bit_Counter.2{P8}
/CPU_$2005_WR{L113}
R29
Inverter
INVERT
Buffer
/Rendering_In_Frame{R24}
/CPU_$2002_RD{L110}
R28
BUFFER
R27
D
Q
/Reset
/Q
D-FF Reset
Resyncs 2nd write tracking
when $2002 is read.
/Q -> D turns this
into a T-FF with
async /reset.

Sheet: /CHR A0,A1,A2/
File: chr_a0_a1_a2.kicad_sch

Title:

Size: USLedger | Date: | **Rev:**
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$5104 Register ExRAM Mode

CPU_D0_IN_PIN
CPU_D1_IN_PIN
/Reset_Detect{BL97}
/CPU_$5104_WR{L37}
BK82
Reg 2 DFF-Set
BK79
BK87
+ExRAM_Mode_%10{BK87}
BK80
BL49
/ExRAM_Mode_%01{BK80}
/ExRAM_Mode_%01{BL49}
BK81
BL96
/ExRAM_Mode_%00{BK81}
Appears to go nowhere.
+ExRAM_Mode_$0x{BL96}

Tracks whether $5104 has been written since reset detection

/Reset_Detect{RAX12}
M2_CLK_BUF1
Syncs SL3 to M2
+Invalid_A0_M2_Sequence{BN1}
BL89
BL90
BL91
BL98
D-FF Reset
BL94
BL92
BL93
BN2
BIGBUF

There appears to be a selectable delay jumper, connected at BM6.

BM10 BM9 BM8 BM7 BM6 BM5 BM4 BM3 BM2 BM1

+ExRAM_Mode_$0x{BL96}
BL86
BL88
BL85
BL84
BL83
PPU_RW_Mode_%0x_Access{BL83}
Not sure about polarity; this may be Mode %1x Access, and active low or high.
+PPU_RD_A_PIN{PinInv}
+PPU_WR_PIN{PinInv}
BL82
BL87
BL81
BIGBUF

+CPU_ExRAM_Access{L117}
+Reg_Write{BK88}
BK85
/CPU_Write_To_ExRAM_Any_Mode{BK85}
BK86
Writing to ExRAM via $5C00-5FFF in mode %10
+ExRAM_Mode_%10{BK87}

Writing to an ExRAM nametable via PPU via $2000

+ExRAM_Nametable_Access{A9}
+ExRAM_Mode_$0x{BL96}
+PPU_WR_PIN{PinInv}
BK83
BK84
+ExRAM_WR{BK84}
This is the write signal for the ExRAM itself.

BN4
BIGBUF
BL105
BL106
BL109
+ExRAM_Mode_$0x{BL96}
/CPU_Write_To_ExRAM_Any_Mode{BK85}
/Q{BL109}
BL108
BL99
/Q{BL99}
+PPU_A13_A{BK97}
BL102
/Reset_Detect{RAX12}
BL101
BL100
BN3
BIGBUF
BL107
+PPU_RD_A_PIN{PinInv}

PPU is fetching from nametable or attribute ($2000+) and V-Split is not rendering and ExRAM mode is %0x (a PPU mode)

+PPU_A13_A{BK97}
+ExRAM_Mode_$0x{BL96}
BK94
BK95
+Drive_ExRAM_Ax_From_PPU_Ax{BK95}
/V-Split_Rendering{N15}

PPU is fetching from nametable or attribute ($2000+) and V-Split is rendering and ExRAM mode is %0x (a PPU mode)

BK92
BK93
BK91
+Drive_ExRAM_Ax_For_Substituted_Fetch{BK91}

PPU is fetching from pattern table (below $2000) or ExRAM mode is %1x (a CPU mode)

BK96
BJ51
+Drive_ExRAM_Ax_From_CPU_Ax{BJ51}

Note:
Maybe N15 means /Extended Attribute Fetch OR V-Split Rendering?
N15 seems to directly depend on V-Split enable register bit though.

PPU_Fetch_Mux_0{M1} PPU_A0_Pin BJ50 ExRAM_A0{BJ50}
PPU_Fetch_Mux_1{M1} PPU_A1_Pin BJ49 ExRAM_A1{BJ49}
PPU_Fetch_Mux_2{M1} PPU_A2_Pin BJ48 ExRAM_A2{BJ48}
PPU_Fetch_Mux_3{M1} PPU_A3_Pin BJ47 ExRAM_A3{BJ47}
PPU_Fetch_Mux_4{M1} PPU_A4_Pin BJ46 ExRAM_A4{BJ46}
PPU_Fetch_Mux_5{M1} PPU_A5_Pin BJ45 ExRAM_A5{BJ45}
PPU_Fetch_Mux_6{M1} PPU_A6_Pin BJ44 ExRAM_A6{BJ44}
PPU_Fetch_Mux_7{M1} PPU_A7_Pin BJ43 ExRAM_A7{BJ43}
PPU_Fetch_Mux_8{M1} PPU_A8_Pin BJ42 ExRAM_A8{BJ42}
PPU_Fetch_Mux_9{M1} PPU_A9_Pin BJ41 ExRAM_A9{BJ41}
NAND Mux 3

CPU_A0_Pin
CPU_A1_Pin
CPU_A2_Pin
CPU_A3_Pin
CPU_A4_Pin
CPU_A5_Pin
CPU_A6_Pin
CPU_A7_Pin
CPU_A8_Pin
CPU_A9_Pin
BJ40
Reg 10 D-Latch

/Q{BL109}
/CPU_Write_To_ExRAM_Any_Mode{BK85}
+ExRAM_Mode_$0x{BL96}
BK89

MMC5 Only
+PPU_RD_A_PIN{PinInv}

MMC5A Only
/Q{BL99}
BK90
+ExRAM_Mode_$0x{BL96}

CPU_D0_IN_PIN … CPU_D7_IN_PIN
BK78
Reg 8 D-Latch
BK77
CPU_D0_In_Latched{BK77} … CPU_D7_In_Latched{BK77}
Reg 8 D-Latch

/Scanline_Counter_Reset{Z25}
+ExRAM_Mode_$0x{BL96}
BL95
BL104
BL103
BJ39b
Inverter
Buffer

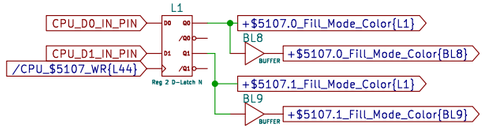
CPU_D0_In_Latched{BK77} PPU_D0_In{BigBuf} BJ39 ExRAM_D0_Write{BJ39}
CPU_D1_In_Latched{BK77} PPU_D1_In{BigBuf} BJ38 ExRAM_D1_Write{BJ38}
CPU_D2_In_Latched{BK77} PPU_D2_In{BigBuf} BJ37 ExRAM_D2_Write{BJ37}
CPU_D3_In_Latched{BK77} PPU_D3_In{BigBuf} BJ36 ExRAM_D3_Write{BJ36}
CPU_D4_In_Latched{BK77} PPU_D4_In{BigBuf} BJ35 ExRAM_D4_Write{BJ35}
CPU_D5_In_Latched{BK77} PPU_D5_In{BigBuf} BJ34 ExRAM_D5_Write{BJ34}
CPU_D6_In_Latched{BK77} PPU_D6_In{BigBuf} BJ33 ExRAM_D6_Write{BJ33}
CPU_D7_In_Latched{BK77} PPU_D7_In{BigBuf} BJ32 ExRAM_D7_Write{BJ32}
NAND Mux 2

ExRAM_D0_Read … ExRAM_D7_Read
BJ12(a) … BJ12(h)
BUFFER
ExRAM_D0_Read{BJ12(a)} … ExRAM_D7_Read{BJ12(h)}
BJ11
/ExRAM_Read_D0_Latch{BJ11} … /ExRAM_Read_D7_Latch{BJ11}
+ExRAM_Read_D6_Latch{BJ11}
+ExRAM_Read_D7_Latch{BJ11}

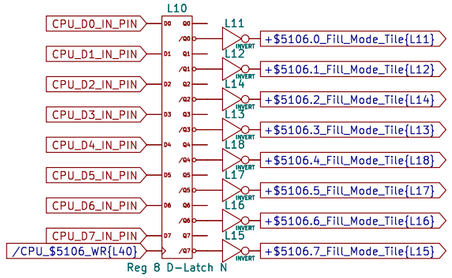
+PPU_RD_A{BJ8}
+PPU_Fetch_Count.1{S13}
+PPU_Fetch_Count.0{S10}
BJ9
BJ10
BJ7
Reg 8 D-Latch N

Sheet: /ExRAM Mode/
File: exram_mode.kicad_sch
Title:
Size: USLedger | Date:
KiCad E.D.A. 9.0.7
Rev: Id: 14/27

$5107 Register
Fill-Mode Color



$5106 Register
Fill-Mode Tile



Sheet: /Fill Mode/
File: fill_mode.kicad_sch

Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 15/27

5-Bit to 7-Bit Pulse Length Lookup Table

See spreadsheet for decoded values: "pulse length lookup table.xlsx"

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
+CPU_D3{E60}
→ D46 (NAND_5) → D45 (INVERT)

+CPU_D6{E63}
+CPU_D5{E57}
+CPU_D4{E56}
/CPU_D3{E55}
→ D48 (NAND_4) → D47 (INVERT)

D45, D47 → C16 (NOR_2) → C15 (INVERT) → Pulse_LUT_D6{C15}

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
+CPU_D4{E61}
/CPU_D3{E55}
→ D50 (NAND_5) → D49 (INVERT)

+CPU_D7{E64}
+CPU_D6{E63}
/CPU_D5{E57}
/CPU_D3{E55}
→ D52 (NAND_4) → D51 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
+CPU_D5{E62}
+CPU_D4{E61}
/CPU_D3{E55}
→ D54 (NAND_5) → D53 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
+CPU_D3{E60}
→ D56 (NAND_5) → D55 (INVERT)

D49, D51, D53, D55 → C18 (NOR_4) → C17 (INVERT) → Pulse_LUT_D5{C17}

+CPU_D7{E64}
+CPU_D6{E63}
+CPU_D5{E62}
/CPU_D4{E61}
/CPU_D3{E55}
→ D57 (NAND_5) → D58 (INVERT)

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D3{E55}
→ D59 (NAND_4) → D60 (INVERT)

/CPU_D7{E59}
+CPU_D6{E63}
/CPU_D5{E57}
+CPU_D4{E61}
/CPU_D3{E55}
→ D61 (NAND_5) → D62 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
/CPU_D3{E55}
→ D63 (NAND_5) → D64 (INVERT)

D58, D60, D62, D64 → C20 (NOR_4) → C19 (INVERT) → Pulse_LUT_D4{C19}

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
+CPU_D3{E60}
→ D65 (NAND_5) → D66 (INVERT)

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D4{E61}
/CPU_D3{E55}
→ D67 (NAND_5) → D68 (INVERT)

/CPU_D7{E59}
/CPU_D4{E61}
/CPU_D3{E55}
→ D69 (NAND_3) → D70 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
+CPU_D3{E60}
→ D71 (NAND_5) → D72 (INVERT)

+CPU_D7{E64}
+CPU_D3{E60}
→ D73 (NAND_2) → D74 (INVERT)

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D4{E56}
→ D75 (NAND_4) → D76 (INVERT)

D68, D70, D72, D74, D76 → C22 (NOR_5) → C21 (INVERT) → Pulse_LUT_D3{C21}

+CPU_D6{E63}
/CPU_D5{E57}
+CPU_D4{E61}
→ D77 (NAND_3) → D78 (INVERT)

/CPU_D7{E59}
+CPU_D6{E63}
+CPU_D5{E62}
/CPU_D4{E56}
/CPU_D3{E55}
→ D79 (NAND_4) → D80 (INVERT)

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D3{E55}
→ D81 (NAND_4) → D82 (INVERT)

/CPU_D7{E59}
+CPU_D6{E63}
+CPU_D5{E62}
→ D83 (NAND_3) → D84 (INVERT)

+CPU_D6{E63}
+CPU_D3{E60}
→ D85 (NAND_2) → D86 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
→ D87 (NAND_4) → D88 (INVERT)

D78, D80, D82, D84, D86, D88 → C24 (NOR_6) → C23 (INVERT) → Pulse_LUT_D2{C23}

+CPU_D7{E64}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
/CPU_D3{E55}
→ D89 (NAND_5) → D90 (INVERT)

/CPU_D7{E59}
+CPU_D6{E63}
/CPU_D5{E57}
+CPU_D4{E61}
/CPU_D3{E55}
→ D91 (NAND_5) → D92 (INVERT)

/CPU_D7{E59}
+CPU_D6{E63}
+CPU_D5{E62}
/CPU_D4{E56}
/CPU_D3{E55}
→ D93 (NAND_5) → D94 (INVERT)

+CPU_D5{E62}
+CPU_D3{E60}
→ D95 (NAND_2) → D96 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
+CPU_D4{E61}
/CPU_D3{E55}
→ D97 (NAND_5) → D98 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D3{E55}
→ D99 (NAND_5) → D100 (INVERT)

D90, D92, D94, D96, D98, D100 → C26 (NOR_6) → C25 (INVERT) → Pulse_LUT_D1{C25}

/CPU_D7{E59}
+CPU_D6{E63}
+CPU_D5{E62}
/CPU_D3{E55}
→ D101 (NAND_4) → D102 (INVERT)

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
→ D103 (NAND_4) → D104 (INVERT)

+CPU_D4{E61}
+CPU_D3{E60}
→ D105 (NAND_2) → D106 (INVERT)

D102, D104, D106 → C28 (NOR_3) → C27 (INVERT) → Pulse_LUT_D0{C27}

$5003 Register Pulse 1 Length Counter

+$5000.5_Length_Halt{G85} → B44 (INVERT) → B43 (NAND_2)
Pulse_1_L_Restart{C31(b)} → B43 (NAND_2)
B43 → B42 (NOR_2)

Pulse_LUT_D0{C27}
Pulse_LUT_D1{C25}
Pulse_LUT_D2{C23}
Pulse_LUT_D3{C21}
→ C11 (P0–P3, Q0–Q3, Ripple Out, /Load, Counter 4 Preload)

Pulse_LUT_D4{C19}
Pulse_LUT_D5{C17}
Pulse_LUT_D6{C15}
+Reset_Detect{D40}
→ C12 (P0–P3, Q0–Q3, Ripple Out, /Load, Counter 4 Preload) → Pulse_1_L_Rollover{C12}

/CPU_$5003_WR{K16} → /Load

These counters count "down" the preloaded period value.

This means either the outputs or preload inputs are inverted.

Pulse_Length_Clock_SL3(Reset_Aware){A75}
241.08Hz

$5007 Register Pulse 2 Length Counter

+$5004.5_Length_Halt{E68} → C8 (INVERT) → C9 (NAND_2)
Pulse_2_L_Restart{C31(a)} → C9 (NAND_2)
C9 → C10 (NOR_2)

Pulse_LUT_D0{C27}
Pulse_LUT_D1{C25}
Pulse_LUT_D2{C23}
Pulse_LUT_D3{C21}
→ D38 (P0–P3, Q0–Q3, Ripple Out, /Load, Counter 4 Preload)

Pulse_LUT_D4{C19}
Pulse_LUT_D5{C17}
Pulse_LUT_D6{C15}
+Reset_Detect{D40}
→ D39 (P0–P3, Q0–Q3, Ripple Out, /Load, Counter 4 Preload) → Pulse_2_L_Rollover{D39}

/CPU_$5007_WR{K80} → /Load

Pulse Channel Clock Source

1.789773 MHz
+M2_CLK_BUF2{B109} → A56 (NOR_2), A57 (NOR_2)

A56 → A60 (INVERT) → A61 (INVERT) → Pulse_Timer_Clock_Fast{A61}
A57 → A58 (INVERT) → A59 (INVERT) → Pulse_Timer_Clock_Fast{A59}
894.886 kHz
<25% duty, depends on M2 Duty

1
/Reset_Detect{A63} → A55 (T, +Reset, /Q, Q; T-FF Reset)
894.886 kHz
50% duty

The reset inputs of ALL these counters seem to be /Reset. Hard to see why this acts different than T-FF counters elsewhere.

A66 (Q0–Q7, +Reset, Counter 8 Reset)
/Reset_Detect{A63} → A66
A65 resets Q7 of A66.

A66 → A66.2 (INVERT)
447.443 kHz
50% duty

3.49565 kHz
50% duty

A66 → A66.3 (INVERT) → A66.6 (INVERT) → A66.7 (INVERT) → A68 (INVERT) → A66.4 (NAND_5)
/Value 3712

A67 (Q0–Q3, +Reset, Counter 5 Reset)
Important: A63 resets Q4 of A67.

/Reset_Detect{A63} → A64 (NAND_2) → A65 (INVERT)
A65 → A67 +Reset

+Reset_Detect{A62} → A66.5 (NOR_3)
A66.5 → A64 (NAND_2)
always 0 when reset detected, without reset, only 0 when value=3712

A66.2 in this logic can only create a phase shift and not affect the period since it doesn't cause A66 to reset.

241.08 Hz
Duty seems thin

A66.4 → A78 (AOI_2)
B108 (INVERT) → 241.08 Hz or Pin 98 Input

A78 → A70 (NAND_2) → A71 (INVERT) → A72 (INVERT)

Pulse_Timer_Clock_SL3{A72}
E51 (INVERT) → /Pulse_Timer_Clock_SL3{E51}
H87 (INVERT) → /Pulse_Timer_Clock_SL3{H87}

All of these outputs are 241.08 Hz –or– Pin 98 Input

A72 → A73 (NAND_2) → A74 (INVERT) → A75 (INVERT) → Pulse_Length_Clock_SL3{Reset_Aware}{A75}

/Q4 is high for the first 240Hz period after reset has stopped being detected.

A83 (INVERT) → B107 (INVERT)

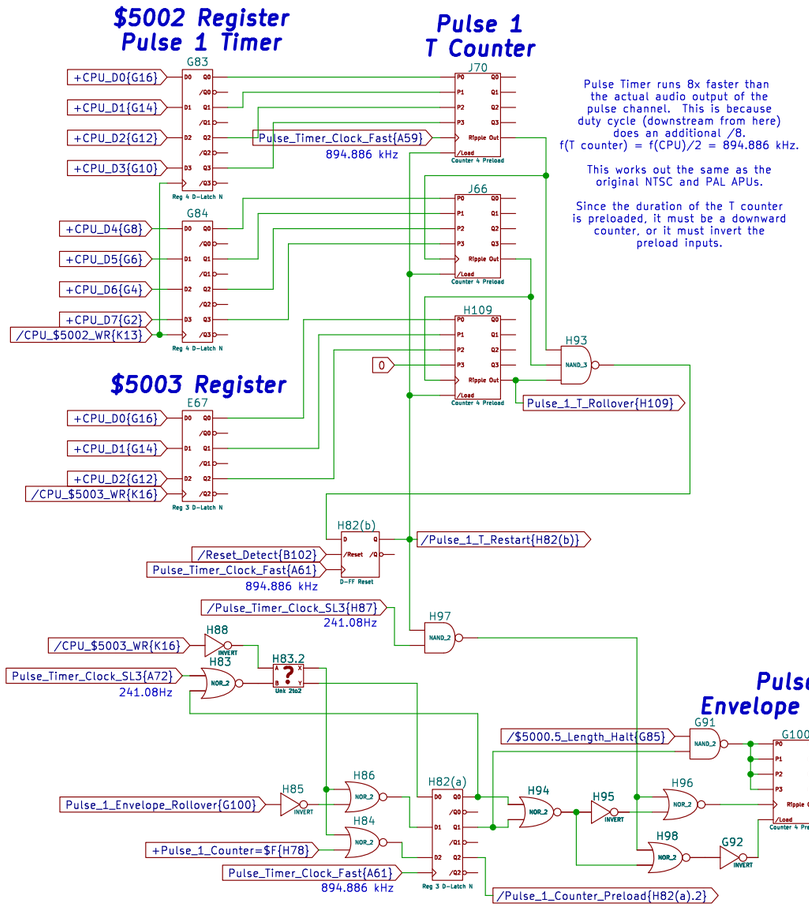
A69 (INVERT)
A79 (INVERT)
A81 (INVERT)
A80 (INVERT)
+Test_Mode_PIN_92{PinInv}
/Test_Input_PIN_30{PinInv}
→ A77 (AOI_2) → A82 (INVERT)

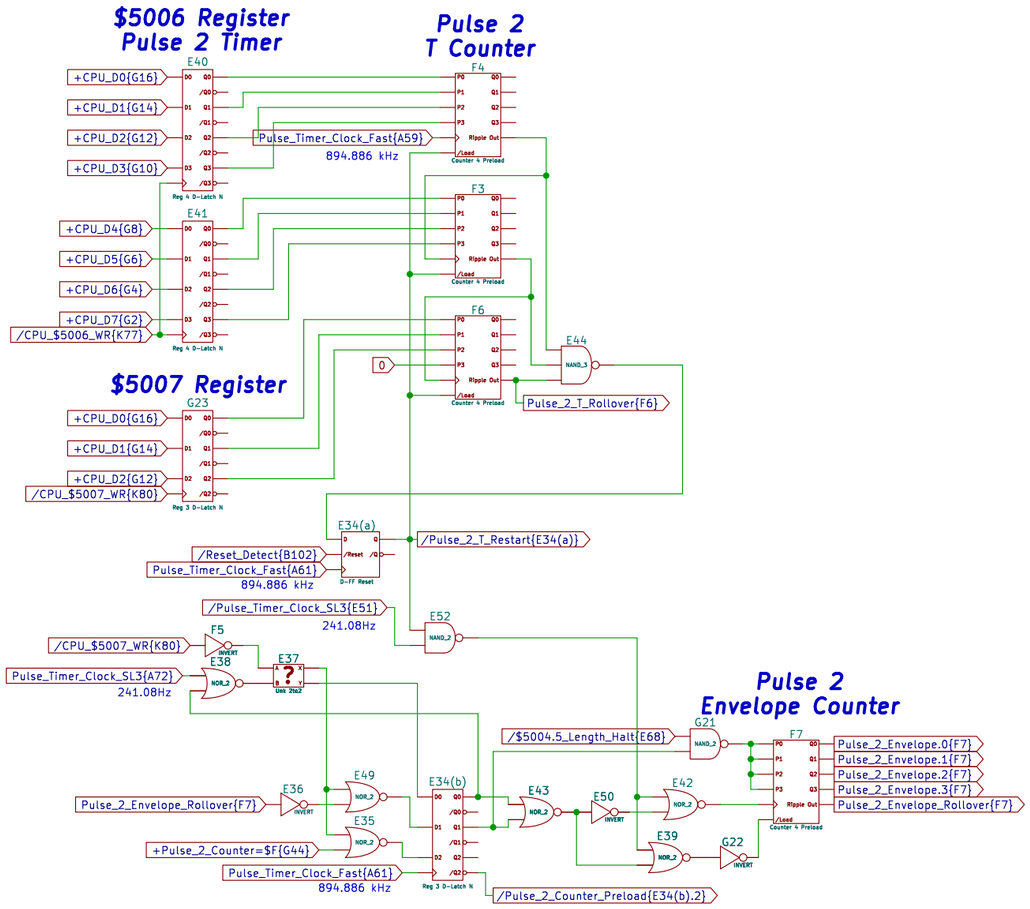
0=Reset Held By A67.Q4 or Pin 30

MMC5 Only
+SL3_PIN{PinBuf}

MMC5A Only
/Invalid_A0_M2_Sequence{BM11}

Sheet: /Pulse Length Lookup Table/	
File: pulse_length_lookup_table.kicad_sch	
Title:	
Size: USLedger Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 16/27



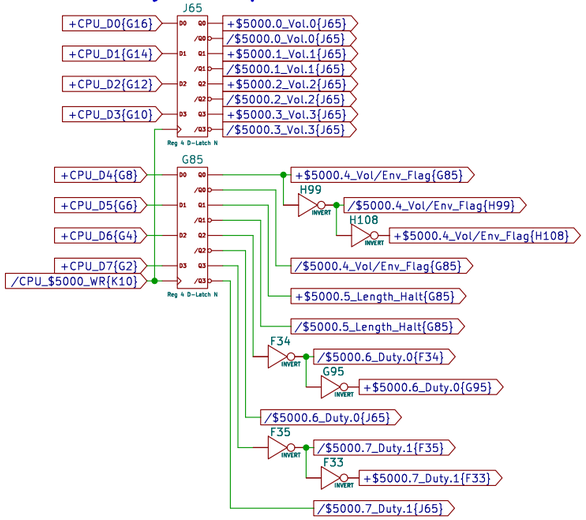


Sheet: /Pulse Timer/
File: pulse_timer.kicad_sch

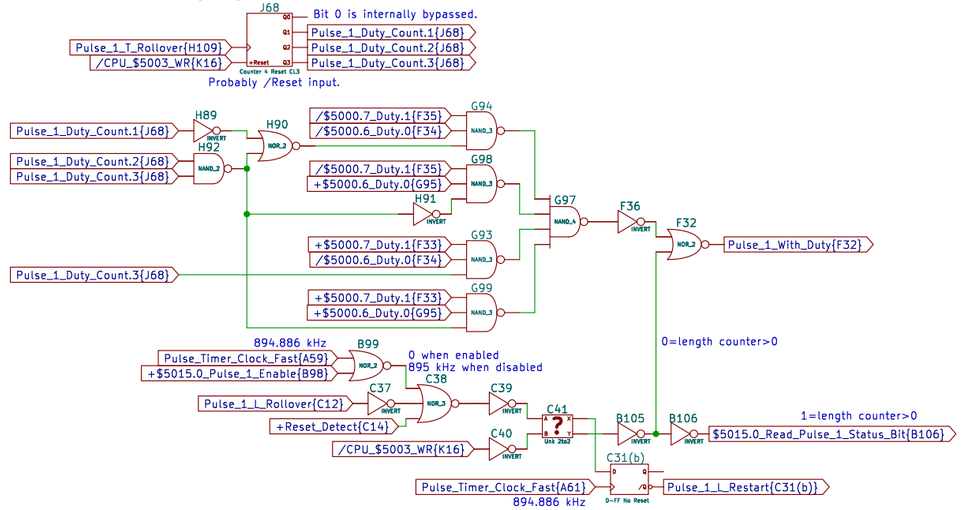
Title:

Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 17/27

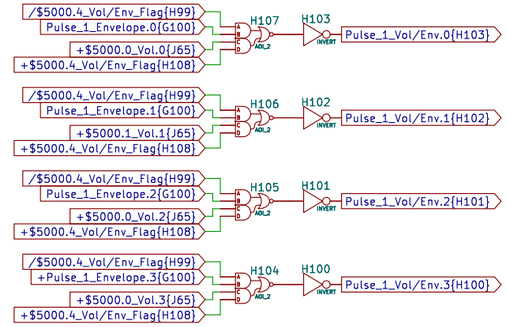
$5000 Register
Pulse 1 Duty Envelope Volume



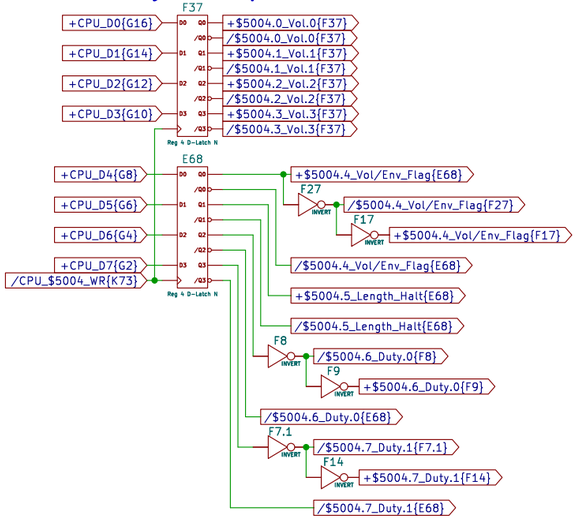
Pulse 1
Duty Cycle Counter



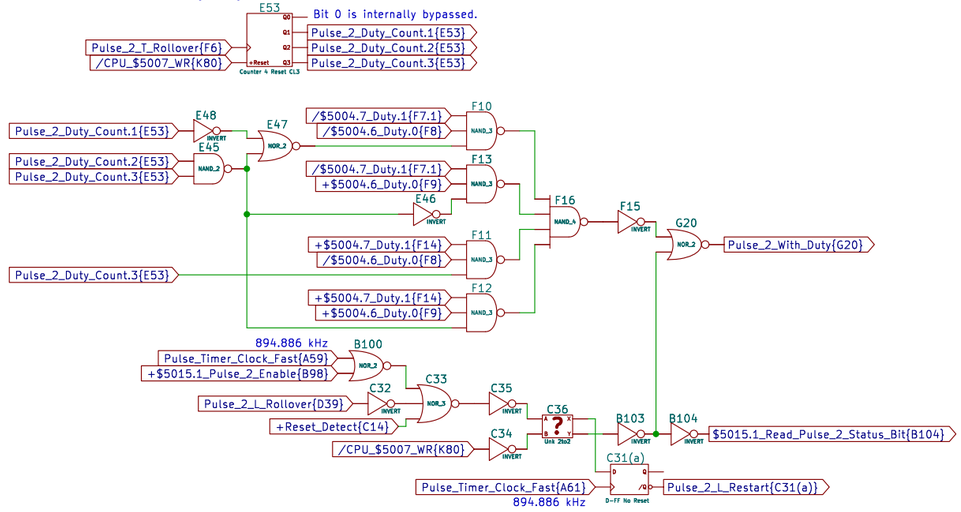
Pulse 1
Envelope/Volume Multiplexing



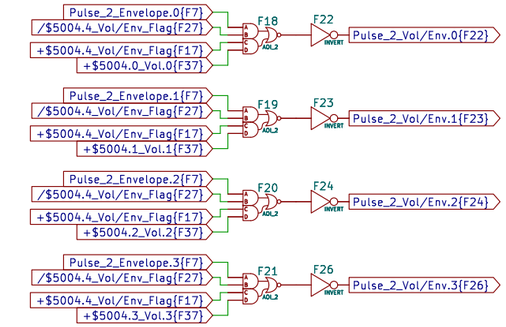
$5004 Register
Pulse 2 Duty Envelope Volume



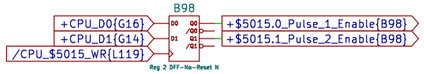
Pulse 2
Duty Cycle Counter



Pulse 2
Envelope/Volume Multiplexing



$5015 Write Register
Pulse 1&2 Status

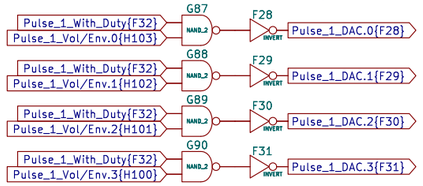


Sheet: /Pulse Duty Envelope Volume/
File: pulse_duty_envelope_volume.kicad_sch

Title:

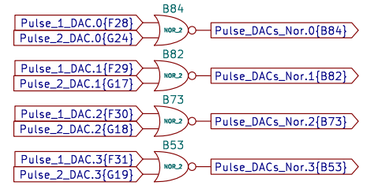
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 18/27

Pulse 1&2
Individual 4-bit DAC Outputs

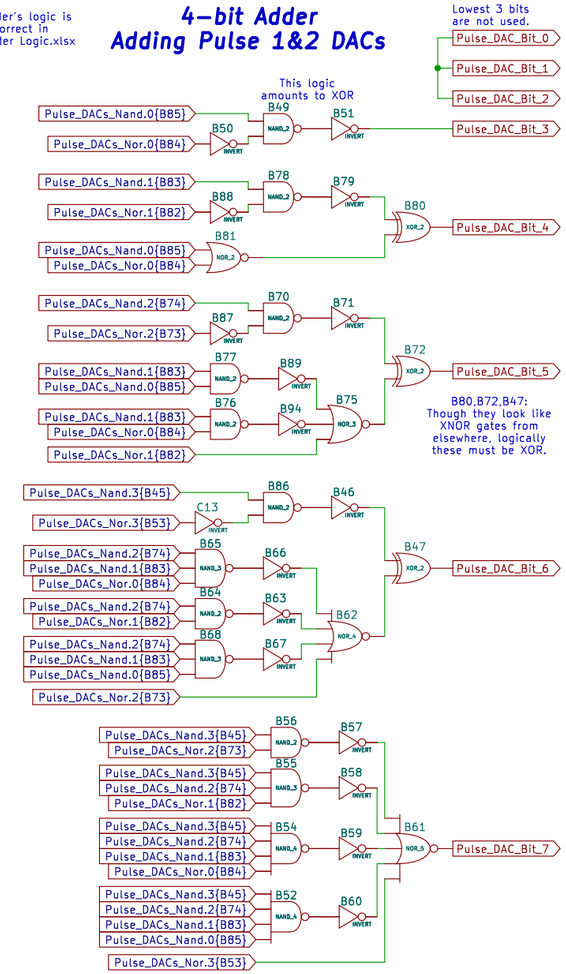


Components for
4-bit Adder

Note: The adder's logic is verified correct in Pulse DAC Adder Logic.xlsx



4-bit Adder
Adding Pulse 1&2 DACs



Sheet: /Pulse DAC/
File: Pulse DAC.kicad_sch

Title:

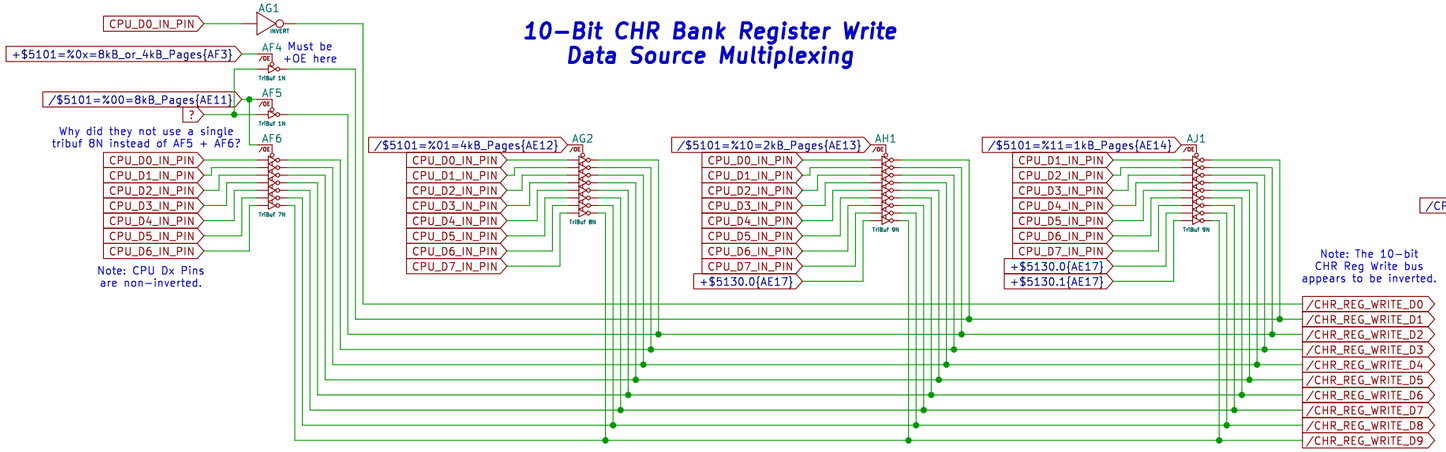
Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 19/27

PPU_D0_IN_PIN
PPU_D1_IN_PIN
PPU_D2_IN_PIN
PPU_D3_IN_PIN
PPU_D4_IN_PIN
PPU_D5_IN_PIN
PPU_D6_IN_PIN
PPU_D7_IN_PIN
{BigBuf}
BIGBUF
PPU_D0_In{BigBuf}
PPU_D1_In{BigBuf}
PPU_D2_In{BigBuf}
PPU_D3_In{BigBuf}
PPU_D4_In{BigBuf}
PPU_D5_In{BigBuf}
PPU_D6_In{BigBuf}
PPU_D7_In{BigBuf}

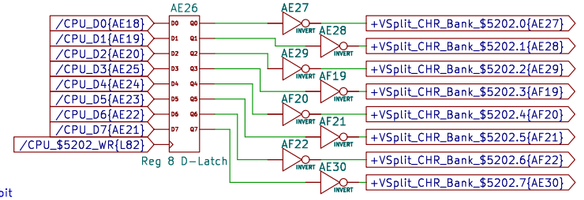
/V-Split_Rendering{BL43}
/Scanline_Counter_Reset{Z25}
/ExRAM_Mode_%01{BL49}
/Extended Attributes Enabled
+Fill_Mode_Nametable_Access{BL33}
/PPU_Attribute_Access{BL25}
+ExRAM_Nametable_Access{A9}
+ExRAM_Mode_$0x{BL48}
BL10
BL34
BL35
BL40
BL32
BL28
BL36
BL37
BL38
BL44
BL41
BL46
BL45
BL42
BL29
BL30
BK39
BL31
BL39
BK40
BK42
BK41
BK43
+$5107.1_Fill_Mode_Color{BL9}
+$5107.0_Fill_Mode_Color{BL8}
BH25
BH27
BH26
+PPU_Fetch_Counter.3{S21}
+V-Split_Scanline_Counter.4{T22}
BH22
BH24
BH20
BH18
BH21
BH23
BH19
BH17
ExRAM_D7_Read{BJ12(h)}
ExRAM_D5_Read{BJ12(f)}
ExRAM_D3_Read{BJ12(d)}
ExRAM_D1_Read{BJ12(b)}
BH4
BH5
BH6
BH7
BH8
BH3
BH2
BH1
+ExRAM_Read_D7_Latch{BJ11}
+ExRAM_Read_D6_Latch{BJ11}
ExRAM_D6_Read{BJ12(g)}
ExRAM_D4_Read{BJ12(e)}
ExRAM_D2_Read{BJ12(c)}
ExRAM_D0_Read{BJ12(a)}
BH12
BH13
BH14
BH15
BH16
BH11
BH10
BH9
ExRAM_D0_Read{BJ12(a)}
+$5106.0_Fill_Mode_Tile{L11}
ExRAM_D1_Read{BJ12(b)}
+$5106.1_Fill_Mode_Tile{L12}
ExRAM_D2_Read{BJ12(c)}
+$5106.2_Fill_Mode_Tile{L14}
ExRAM_D3_Read{BJ12(d)}
+$5106.3_Fill_Mode_Tile{L13}
ExRAM_D4_Read{BJ12(e)}
+$5106.4_Fill_Mode_Tile{L18}
ExRAM_D5_Read{BJ12(f)}
+$5106.5_Fill_Mode_Tile{L17}
ExRAM_D6_Read{BJ12(g)}
+$5106.6_Fill_Mode_Tile{L16}
ExRAM_D7_Read{BJ12(h)}
+$5106.7_Fill_Mode_Tile{L15}
BK1
BL12
BL1
BL13
BL2
BL14
BL3
BL15
BL4
BL16
BL5
BL17
BL6
BL18
BL7
BL19
BK7
BK11
BK15
BK19
BK23
BK27
BK31
BK35
BK10
BK8
BK14
BK12
BK18
BK16
BK22
BK20
BK26
BK24
BK30
BK28
BK34
BK32
BK38
BK36
BK9
BK13
BK17
BK21
BK25
BK29
BK33
BK37
PPU_D0_OUT_PIN
PPU_D1_OUT_PIN
PPU_D2_OUT_PIN
PPU_D3_OUT_PIN
PPU_D4_OUT_PIN
PPU_D5_OUT_PIN
PPU_D6_OUT_PIN
PPU_D7_OUT_PIN
CIRAM_CE_PIN{A9}
+PPU_RD_A{BJ1}
+PPU_A13_B{BK6}
/PPU_A12_B{BK3}
BK2
BK5
BK4
PPU_Dx_Direction{BK4}

Sheet: /PPU Data Bus/
File: ppu_data_bus.kicad_sch
Title:
Size: USLedger | Date: | Rev:
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10-Bit CHR Bank Register Write Data Source Multiplexing



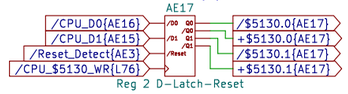
$5202 Register V-Split CHR Bank



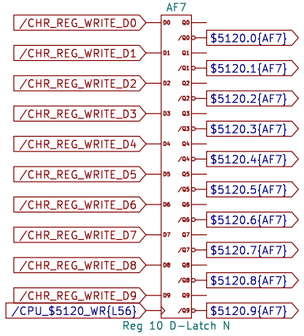
CHR Reg Write Buffering



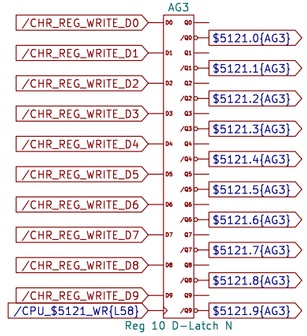
$5130 Register CHR Bank Upper Bits



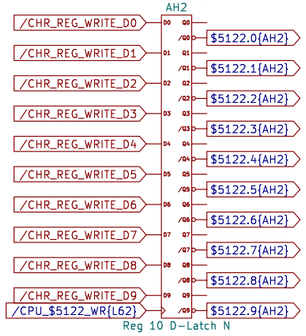
$5120 Register



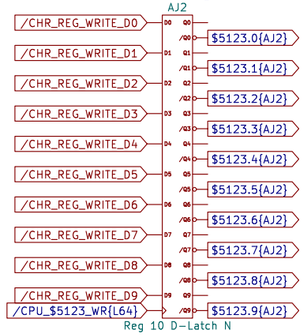
$5121 Register



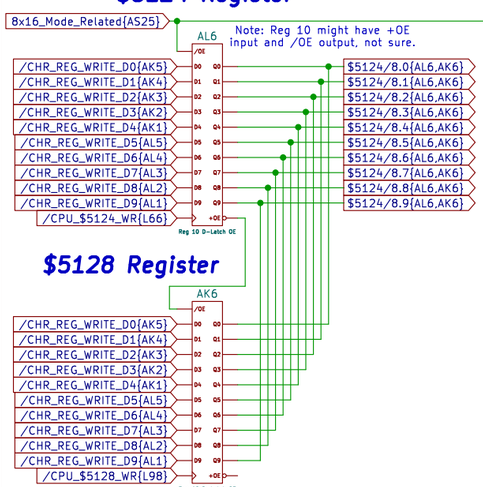
$5122 Register



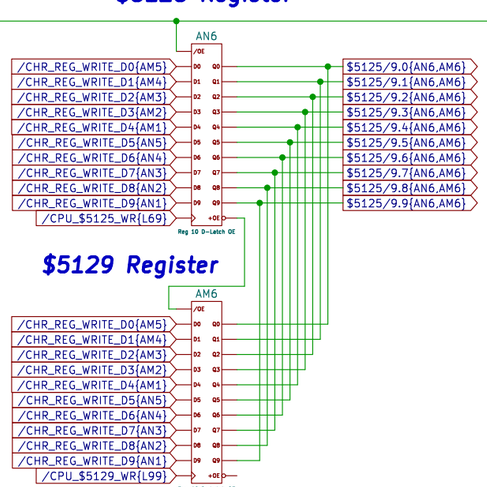
$5123 Register



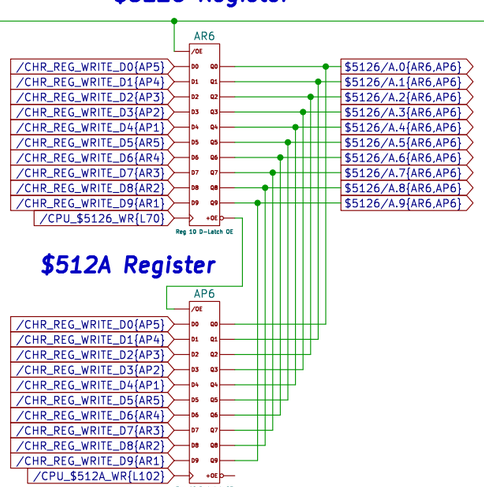
$5124 Register



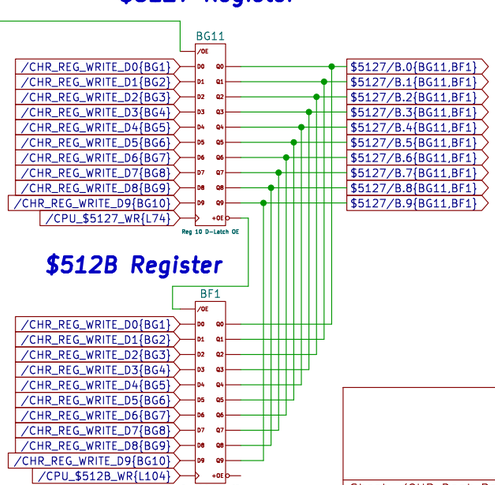
$5125 Register



$5126 Register



$5127 Register



$5128 Register

$5129 Register

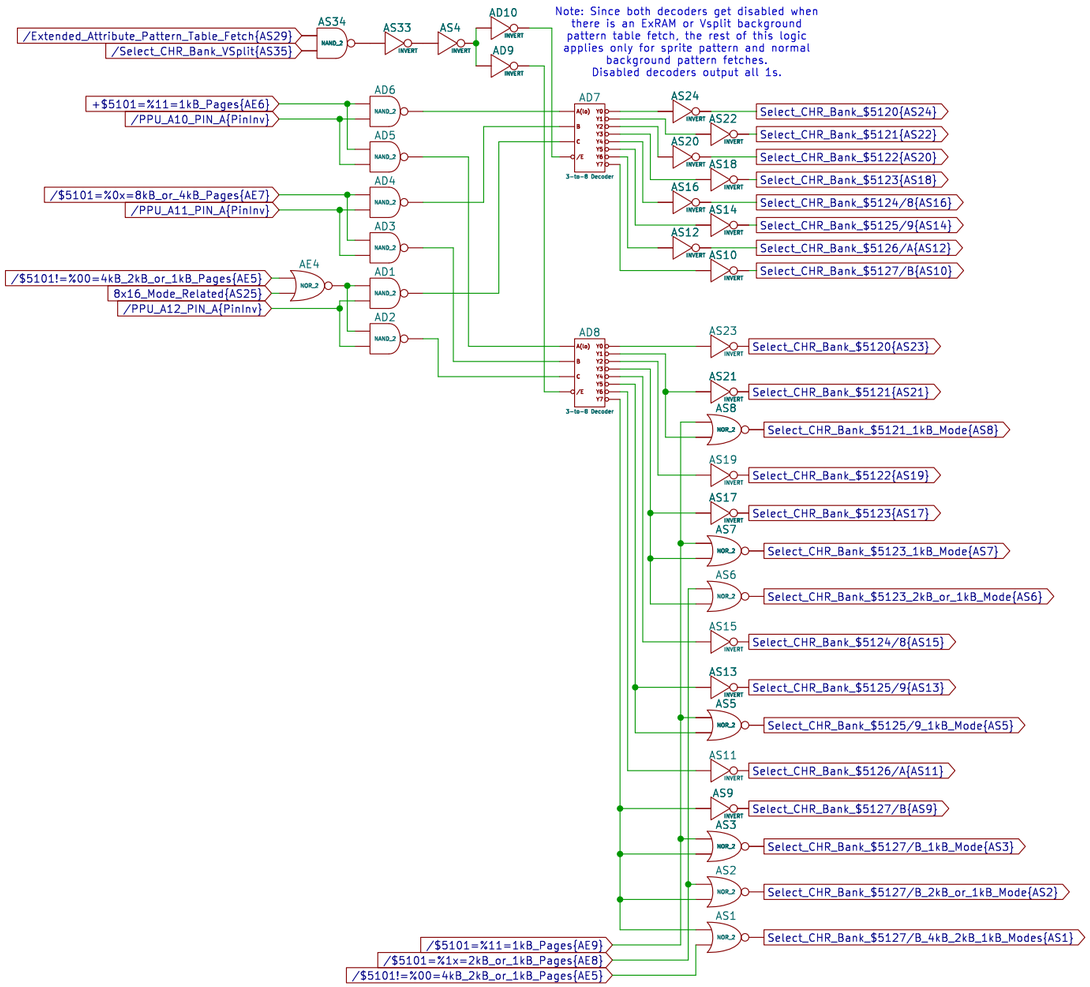
$512A Register

$512B Register

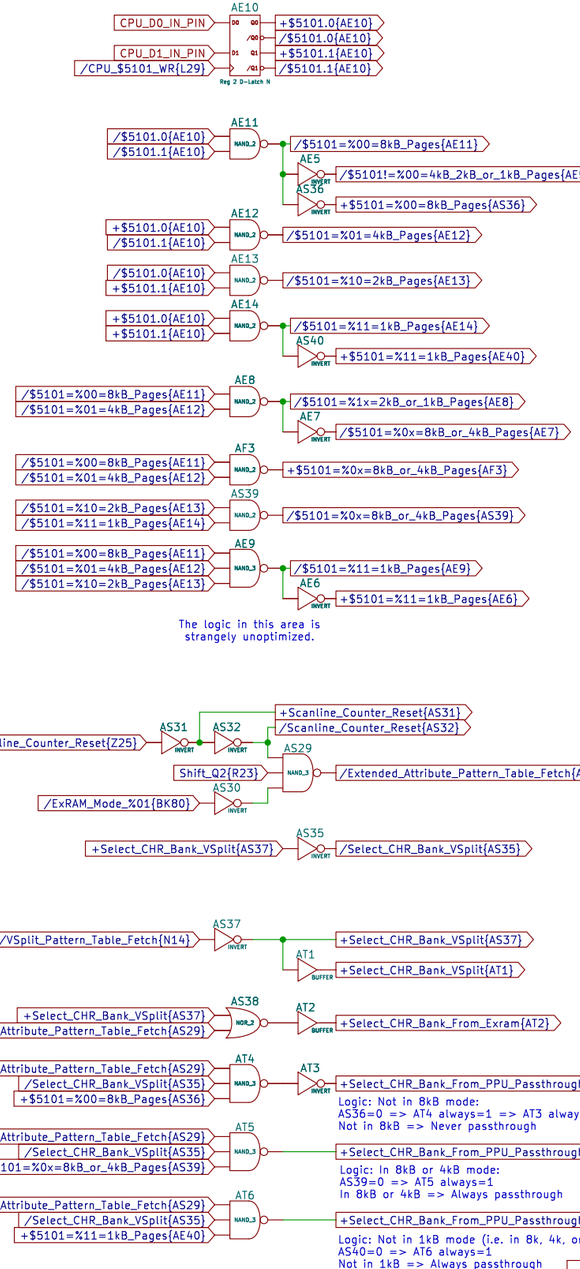
Sheet: /CHR Bank Registers/
File: chr_bank_registers.kicad_sch

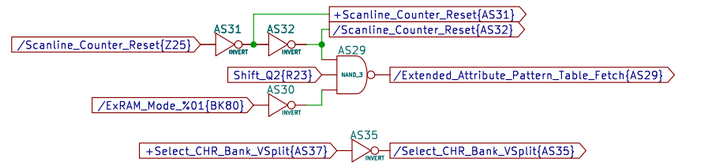
Title:

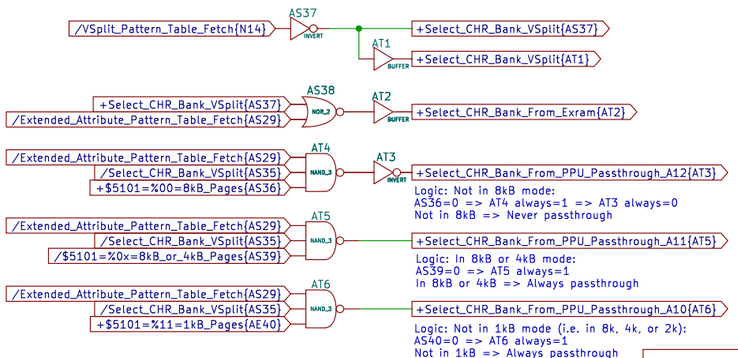
Size: USLedger | Date: | **Rev:**
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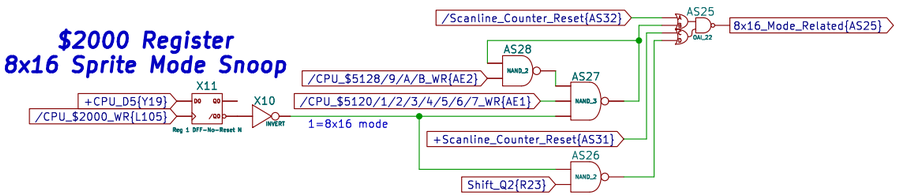
$5101 Register CHR Mode



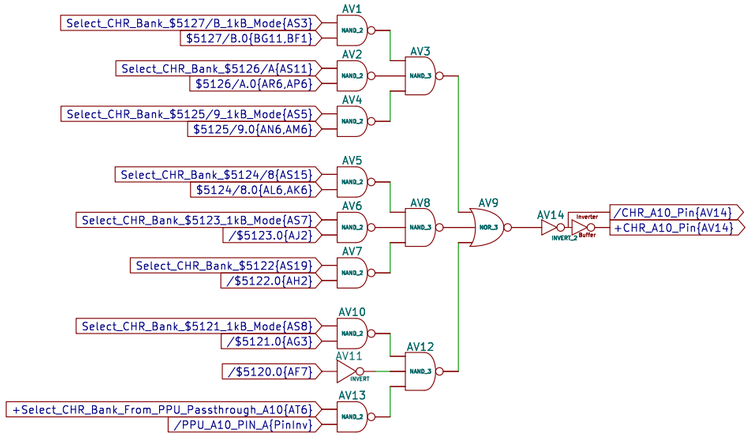




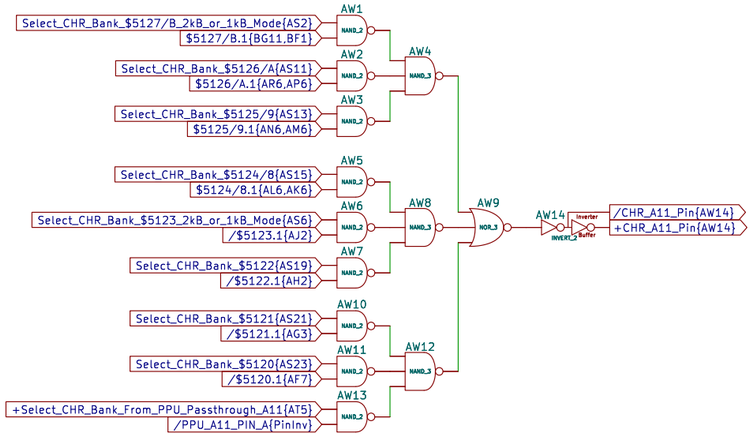
$2000 Register 8x16 Sprite Mode Snoop



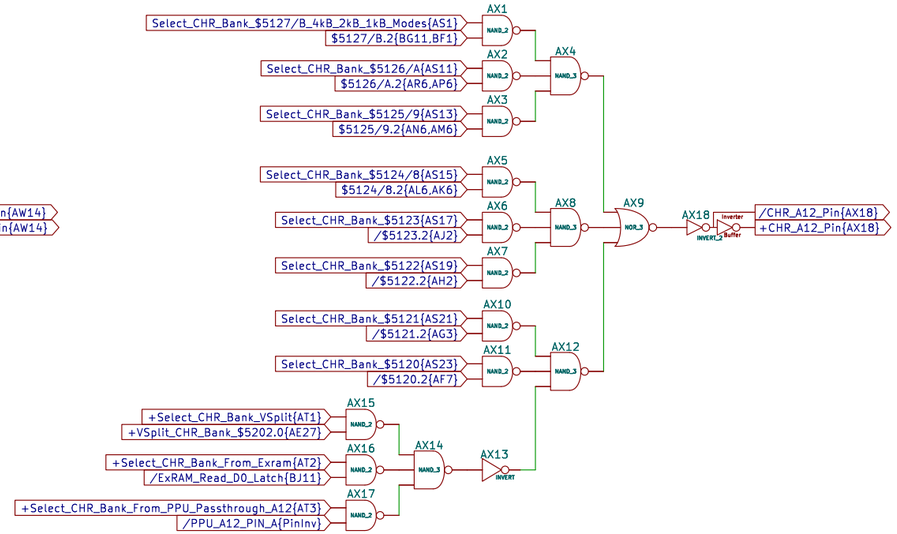
CHR A10



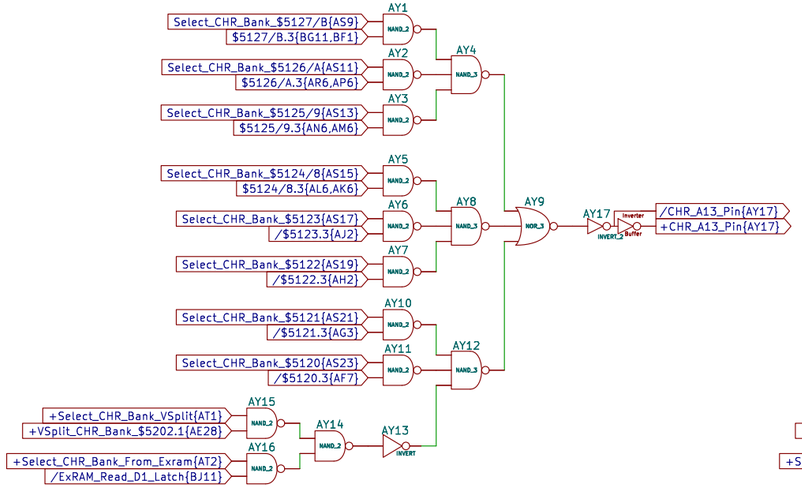
CHR A11



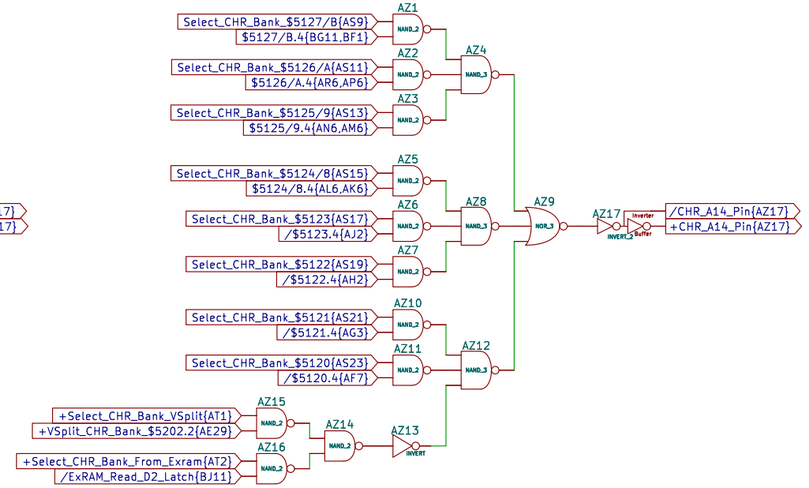
CHR A12



CHR A13



CHR A14

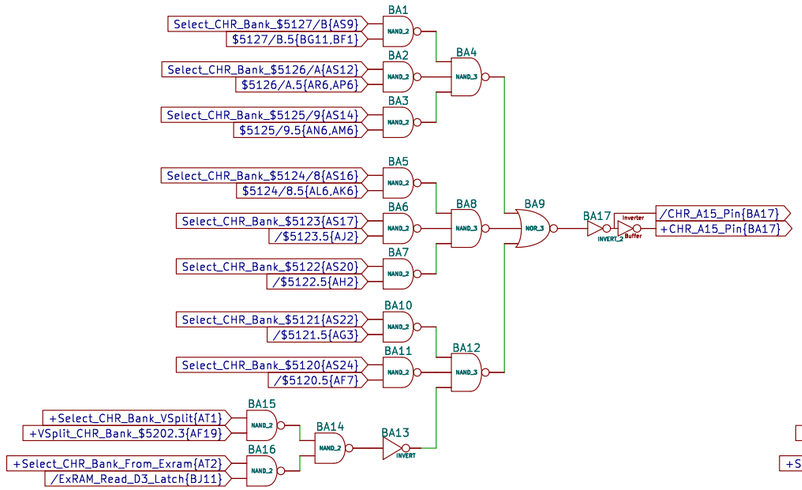


Sheet: /CHR Address Bus 10 to 14/
File: chr_address_bus_10_to_14.kicad_sch

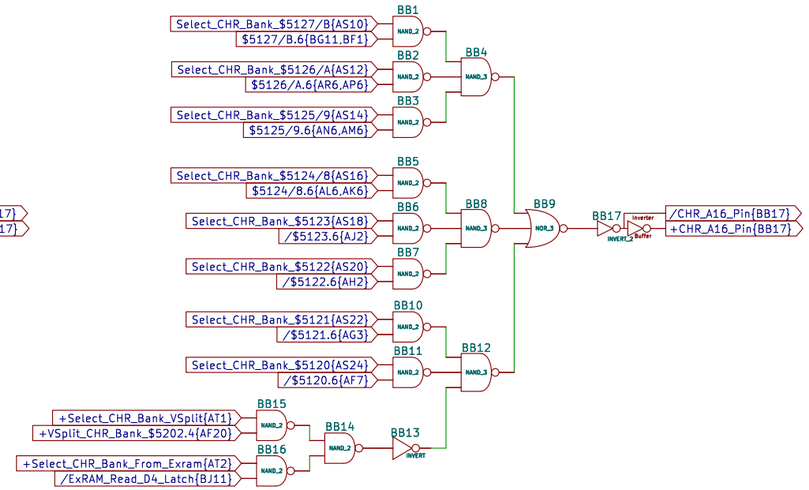
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Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 23/27

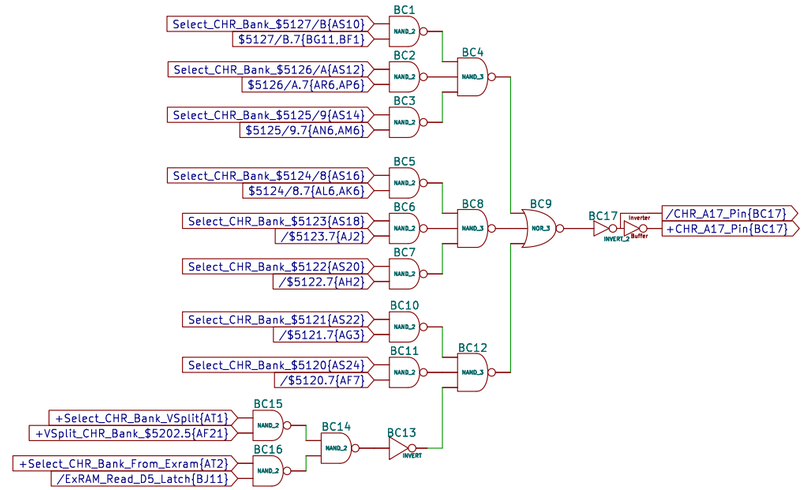
CHR A15



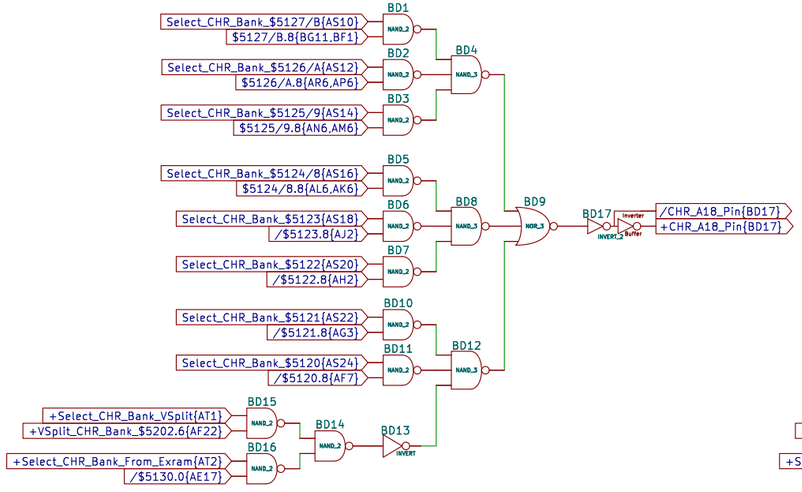
CHR A16



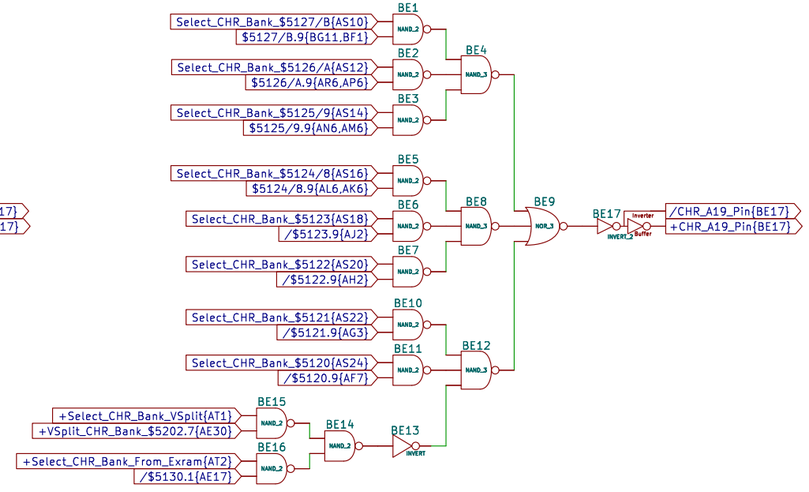
CHR A17



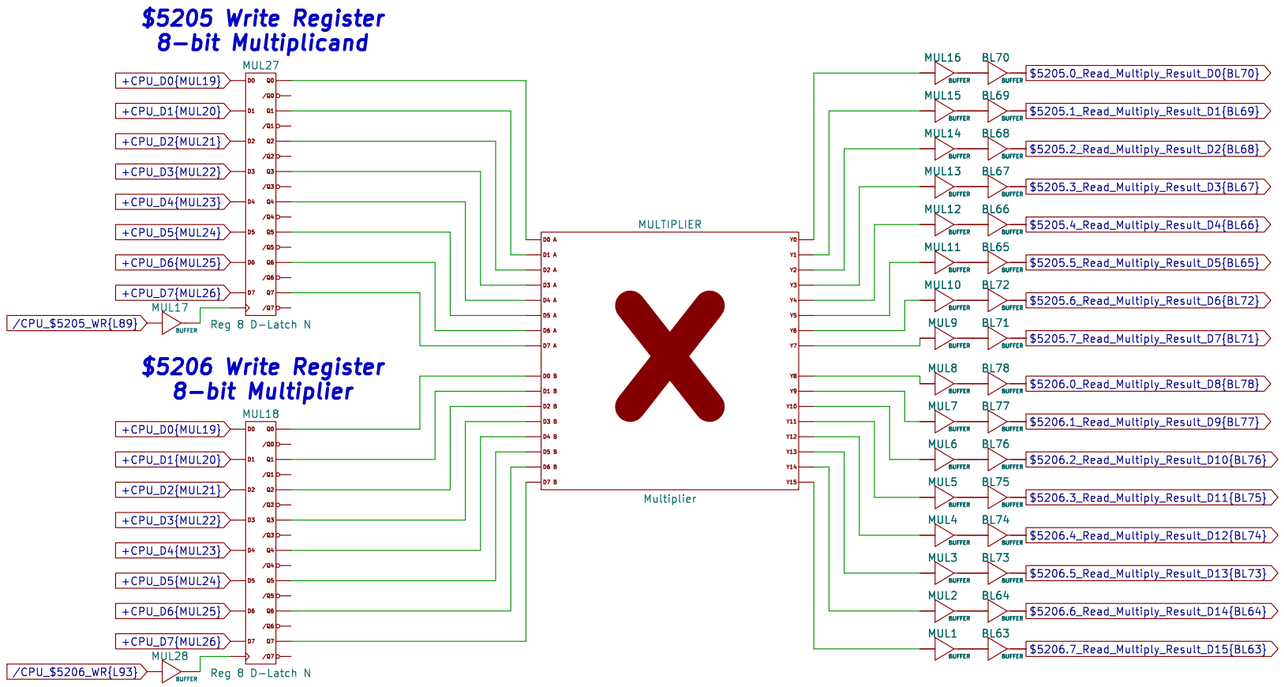
CHR A18



CHR A19



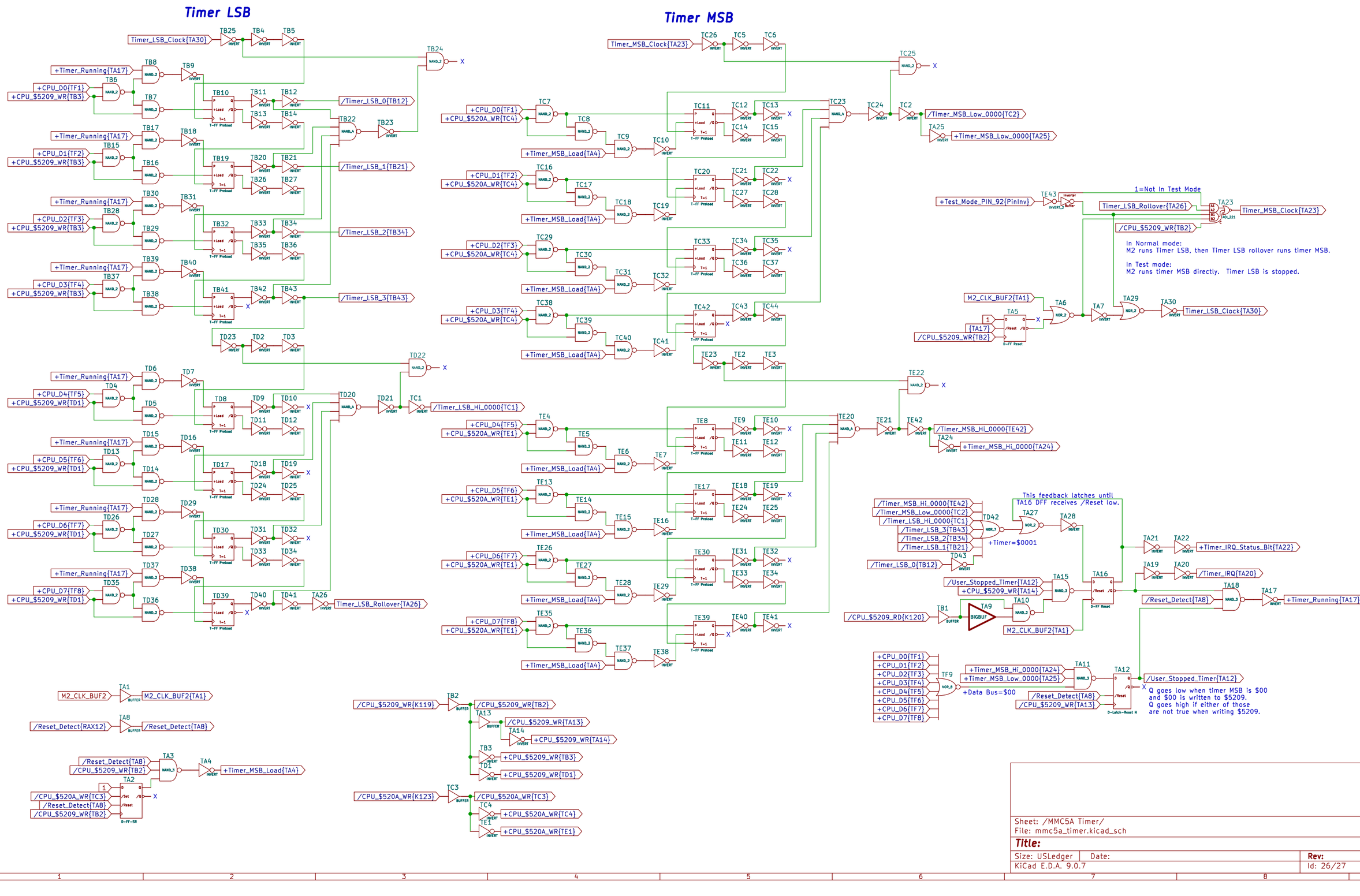
Sheet: /CHR Address Bus 15 to 19/		
File: chr_address_bus_15_to_19.kicad_sch		
Title:		
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 24/27



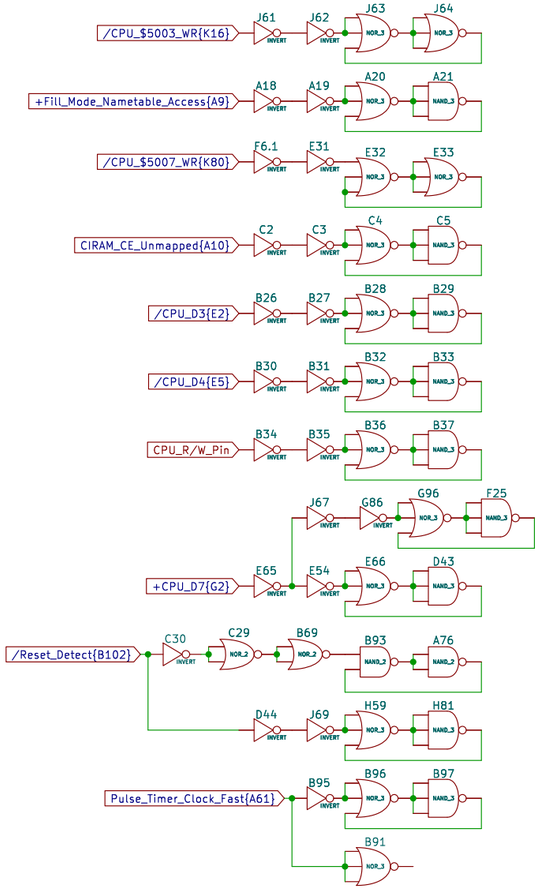
Sheet: /Multiplier/
File: multiplier.kicad_sch

Title:

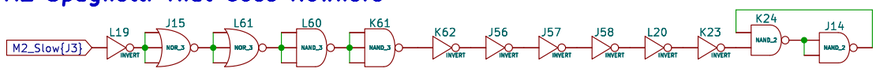
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 25/27

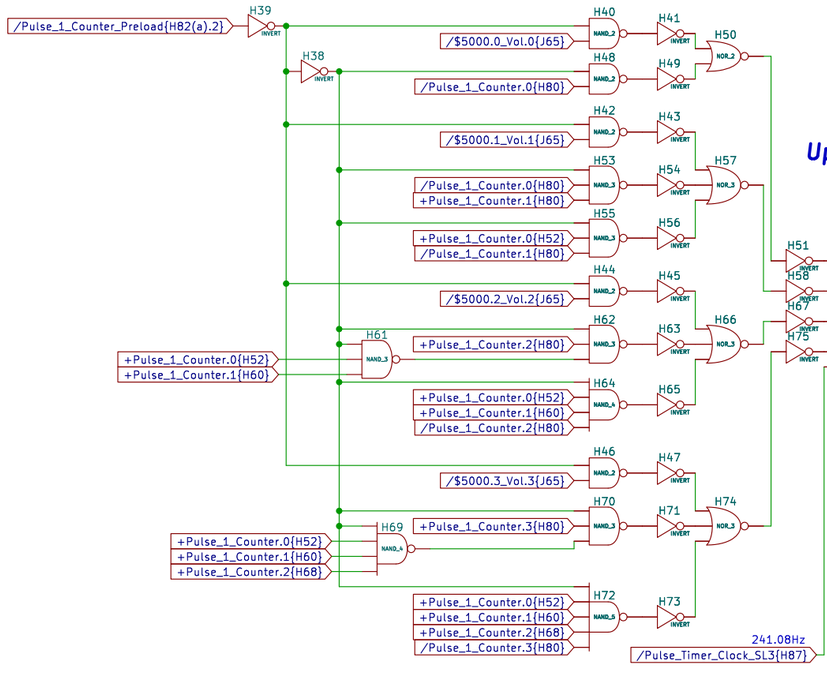

Timer LSB
Timer MSB
Timer_LSB_Clock{TA30}
Timer_MSB_Clock{TA23}
/Timer_LSB_0{TB12}
/Timer_LSB_1{TB21}
/Timer_LSB_2{TB34}
/Timer_LSB_3{TB43}
Timer_LSB_Rollover{TA26}
/Timer_LSB_Hi_0000{TC1}
/Timer_MSB_Low_0000{TC2}
+Timer_MSB_Low_0000{TA25}
/Timer_MSB_Hi_0000{TE42}
+Timer_MSB_Hi_0000{TA24}
+Test_Mode_PIN_92{PinInv}
1=Not In Test Mode
In Normal mode:
M2 runs Timer LSB, then Timer LSB rollover runs timer MSB.
In Test mode:
M2 runs timer MSB directly. Timer LSB is stopped.
This feedback latches until
TA16 DFF receives /Reset low.
+Timer=$0001
+Data Bus=$00
Q goes low when timer MSB is $00
and $00 is written to $5209.
Q goes high if either of those
are not true when writing $5209.
/CPU_$5209_RD{K120}
/CPU_$5209_WR{TB2}
/CPU_$520A_WR{TC3}
/CPU_$520A_WR{TC4}
/CPU_$520A_WR{TE1}
/CPU_$5209_WR{TB3}
/CPU_$5209_WR{TD1}
/CPU_$5209_WR{K119}
/CPU_$520A_WR{K123}
/Reset_Detect{RAX12}
/Reset_Detect{TA8}
/User_Stopped_Timer{TA12}
/CPU_$5209_WR{TA13}
/CPU_$5209_WR{TA14}
/CPU_$5209_WR{TA14}
+Timer_IRQ_Status_Bit{TA22}
/Timer_IRQ{TA20}
+Timer_Running{TA17}
+Timer_MSB_Load{TA4}
M2_CLK_BUF2{TA1}
Sheet: /MMC5A Timer/
File: mmc5a_timer.kicad_sch
Title:
Size: USLedger
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 26/27

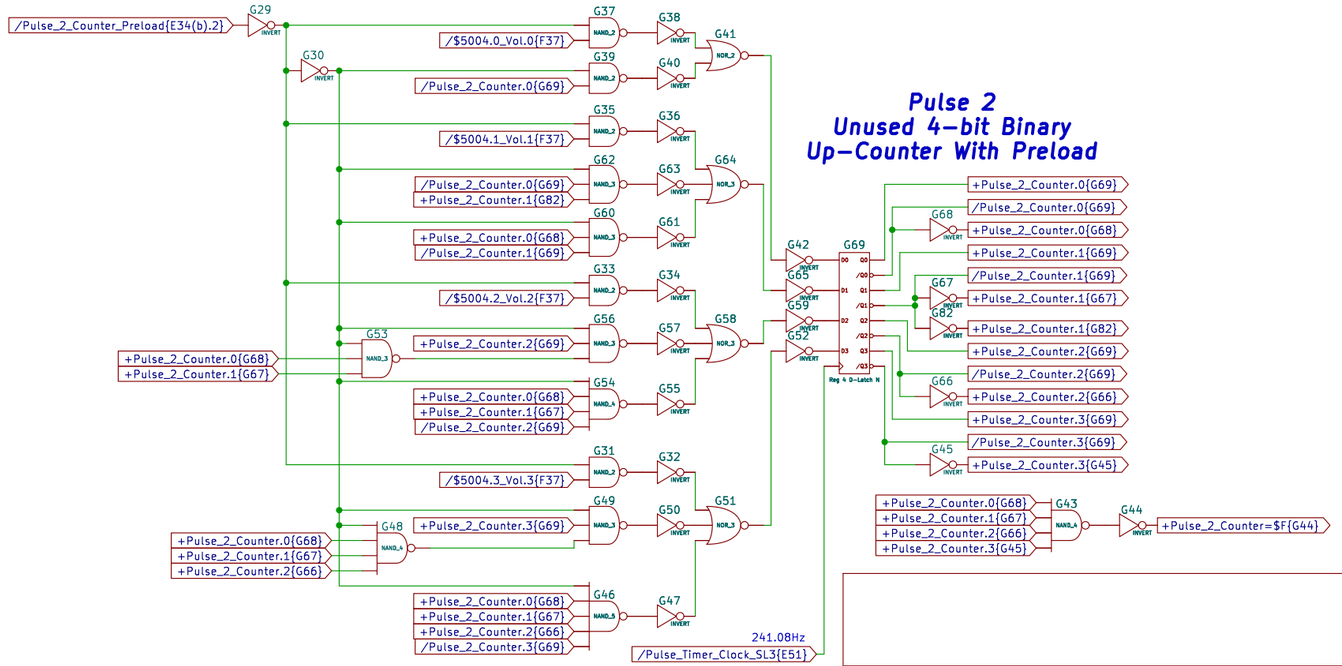
This sheet contains all residual items that don't serve any apparent purpose.



M2 Spaghetti That Goes Nowhere







Pulse 1 Unused 4-bit Binary Up-Counter With Preload

Pulse 2 Unused 4-bit Binary Up-Counter With Preload

Sheet: /Artifacts/
File: artifacts.kicad_sch

Title:

Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 27/27